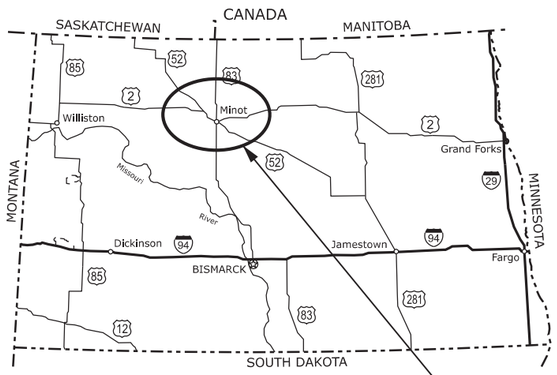
PROJECT	SHEET NUMBER
ND OMAD 2025(1) MINUTEMAN MISSILE ACCESS ROADS	A1



NORTH DAKOTA KEY MAP

U. S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION

PLANS FOR PROPOSED PROJECT

ND OMAD 2025(1)

MINUTEMAN MISSILE ACCESS ROADS

BOTTINEAU, MCHENRY, MCLEAN, WARD, COUNTIES

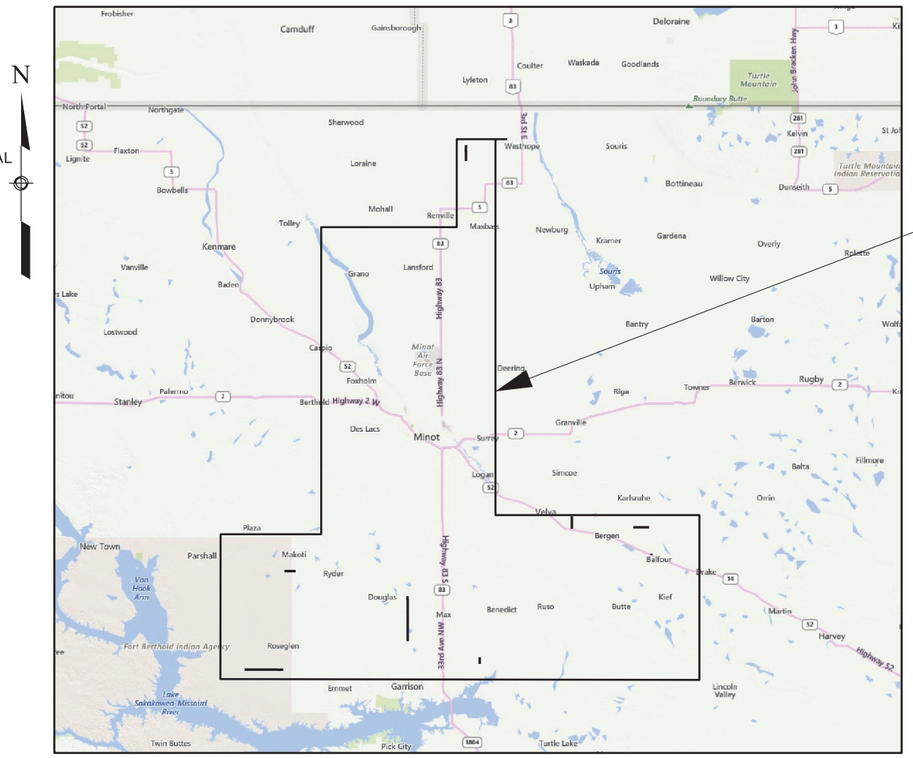
SCHEDULE A LENGTH 9.31 MILES
SCHEDULE B LENGTH 15.52 MILES
SCHEDULE C LENGTH 20.01 MILES

INDEX TO SHEETS

SHEET NUMBER	DESCRIPTION
A1	TITLE SHEET
A2 - A3	CONVENTIONAL PLAN SYMBOLS AND ABBREVIATIONS
A4	KEY MAP
A5	TYPICAL SECTION
B1 - B3	SUMMARY OF QUANTITIES
B4	ROUTE IMPROVEMENT SUMMARY
B5	SURFACING SUMMARY
B6	DRAINAGE SUMMARY
C1 - C10	PLAN SHEETS
T1 - T6	602 STANDARDS - CULVERTS
T7	SPECIAL 633-A OBJECT MARKER DETAIL
T8 - T10	635 STANDARDS - TEMPORARY TRAFFIC CONTROL

TYPE OF CONSTRUCTION:
Aggregate surface course,
aggregate surface reconditioning,
drainage improvements, signage

SPECIFICATIONS:
"STANDARD SPECIFICATIONS FOR CONSTRUCTION OF ROADS AND BRIDGES ON FEDERAL HIGHWAY PROJECTS, FP-24"



Project Location
ND OMAD 2025(1)

PLANS PREPARED BY:

AtkinsRéalis

FOR

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
CENTRAL FEDERAL LANDS HIGHWAY DIVISION
DENVER, CO

REGISTERED PROFESSIONAL ENGINEER
CODY A. BURCH
PE-42098
DATE 08/27/2026
NORTH DAKOTA

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APPROVED:

CURTIS R SCOTT Digitally signed by CURTIS R SCOTT
Date: 2026.09.08 10:19:36 -06'00'

Chief of Engineering, Central Federal Lands Highway Division — *DATE*

Approval per letter signed 8/20/2026
Minot Air Force Base — *DATE*

PROJECT MANAGER
N. MASCARENAS

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18 August 2026 9:11 AM

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	℄	centerline
	Δ	curve delta
	Ø	diameter
A	abut.	abutment
	ADT	average daily traffic
	aggr.	aggregate
	AH	ahead
	alt.	alternate
	appr.	approach
	asph.	asphalt
B	b.f.	both faces
	beg.	beginning, begin
	BK	back
	BM	bench mark
	BP	balance point
	br.	bridge
	brg.	bearing
C	CBC	concrete box culvert
	c-c	center to center
	clr.	clear
	CMP	corrugated metal pipe
	Co.	county
	col.	column
	conc.	concrete
	constr.	construction
	constr. jt.	construction joint
	cont.	continuous
	corr.	corrugated
	cr.	creek
	CS	point of curve to spiral
	ctrs.	centers
	CTSM	contingent sum
	culv.	culvert
D	decr.	decrement
	DHV	design hour volume
	DI	drop inlet
	dia. or D	diameter
	diag.	diagonal
	diaph.	diaphragm
	dist.	distance
	Dist.	district
	DLC	donation land claim
	dwg(s).	drawing(s)
E	E	east
	e	superelevation rate
	El. 94.066	elevation with number
	elev.	elevation
	emb.	embankment
	engr(s).	Engineer(s)
	EOP	edge of pavement
	EQ or eq.	equation
	ER	edge of road
	et al	and others
	et ux	and wife
	EW	edge of water
	exc.	excavation
	exp. jt.	expansion joint
	ext.	exterior
F	f.f.	fill face
	Fed.	federal
	FES	flared end section
	fin.	finish
	ftg.	footing
G	ga.	gage (gauge)
	galv.	galvanized
	gdr.	girder

H	hdwl.	headwall
	HES	homestead entry survey
	hex.	hexagon
	horiz.	horizontal
	HW	high water
	hwy.	highway
I	ID	inside diameter
	incl.	inclusive, including
	int.	interior
J	jt.	joint
L	L	length of curve
	lam.	lamination
	lat.	latitude
	long.	longitude
	LPSM	lump sum
	Lt. or LT	left
	LW	low water
M	mag.	magnetic
	maint.	maintenance
	matl.	material
	max.	maximum
	MGS	Midwest Guardrail System
	min.	minimum
	mon.	monument
	mtn(s).	mountain(s)
N	N	north
	NC	normal crown
	neg.	negative
	no. or #	number
O	o.c.	on centers
	o.f.	other face
	OD	outside diameter
P	PC	point of curve
	PCC	point of compound curve
	perf.	perforate
	PI	point of intersection
	pl.	plate
	POC	point on curve
	POS	point on spiral
	POT	point on tangent
	proj.	project
	psi	pounds per square inch
	PT	point of tangent
	pvmt.	pavement
Q	quant., Qty	quantities
R	R	radius
	R.	range
	R/W	right-of-way
	rd.	road
	rdwy.	roadway
	reconst.	reconstruction
	reinf.	reinforcement
	reqd.	required
	res.	reservoir
	Res.	Reservation
	ret. wall	retaining wall
	RH	reference hub
	Rt. or RT	right
	rte.	route

S	S	south
	SADT	seasonal average daily traffic
	SC	point of spiral to curve
	sec.	section
	shldr.	shoulder
	spa.	spacing, spaces or spaced
	spec.	specification
	st.	street
	ST	point of spiral to tangent
	sta.	station
	std.	standard
	stiff.	stiffener
	str.	straight
	struc.	structural
	sym.	symmetrical
T	T	tangent length
	T.	township
	tan.	tangent
	TBM	temporary bench mark
	TCE	temporary construction easement
	transv.	transverse
	TS	point of tangent to spiral
	typ.	typical
V	V	design speed
	vert.	vertical
	vph	vehicles per hour
	VPI	vertical point of intersection
W	W	West

DRAINAGE SYMBOLS

Ditch (existing, proposed)

Flow Arrow

Drainage or Small Creek

Lake, Pond or Reservoir

Large Creek

Wetland

River

Spring

Bridge (existing, proposed)

Box culvert (existing, proposed)

Pipe culvert (existing, proposed)

With end sections (existing, proposed)

With headwalls (existing, proposed)

With drop inlet (existing, proposed)

Underdrain (existing, proposed)

Riprap apron (proposed)

PROJECT	SHEET NUMBER
ND OMAD 2025(1) MINUTEMAN MISSILE ACCESS ROADS	A2

REGISTERED PROFESSIONAL ENGINEER
CODY A. BURCH
PE-42098
DATE 08/27/2026
NORTH DAKOTA

CONVENTIONAL PLAN SYMBOLS AND ABBREVIATIONS

Sheet 1 of 2

...\SYM\[MAIN]_ND OMAD 2025(1).dgn [Design Intl\Ft]

7/21/2026

PROJECT	SHEET NUMBER
ND OMAD 2025(1) MINUTEMAN MISSILE ACCESS ROADS	A3

FENCE & CATTLE GUARD SYMBOLS

- Fence (existing, proposed)
- Fence with gate (existing, proposed)
- Cattle guard (existing, proposed)

GEOLOGIC SYMBOLS

- Boring location — BH
- Material source

LANDSCAPING & VEGETATION SYMBOLS

- Tree
- Tree line

MAPPING SYMBOLS

- Building (existing, proposed)
- Coordinate Grid Tick
- North arrow — N
- Railroad
  - Single Track
  - Double Track
- Spot elevation — EL. 0.00
- Trail
- Survey control point

GUARDRAIL, BARRIER & WALL SYMBOLS

- Guardrail (existing, proposed)
- Guardwall (proposed)
- Median & Side Barrier (existing, proposed)
- Retaining Wall (existing, proposed)

ROADWAY SYMBOLS

- Clearing/construction limits
- Slope stake limits
  - Top of Cut
  - Transition
  - Toe of fill
- Edge of roadway (existing, proposed)
- Roadway centerline (with station ticks)
- Roadway obliteration

RIGHT-OF-WAY SYMBOLS

- Boundaries
  - National
  - State
  - County
  - City
  - Township or Range line
  - Section
  - 1/4 Section
  - 1/16 Section
  - Bureau of Indian Affairs
  - Bureau of Land Management
  - National Forest
  - National Park — NP
  - National Wildlife Refuge — NWR
- Easements
  - Permanent (existing) — P/E
  - Permanent (proposed) — P/E
  - Temporary (proposed) — TCE
- Monument (as described)
- Parcel number — 400
- Property line — P/L
- Right-of-way line (existing) — r/w
- Right-of-way line (proposed) — R/W
- Section corner (found, projected)
  - 1/4 Section corner (found, projected)
  - 1/16 Section corner (found, projected)

SIGN SYMBOLS

- Signs
  - Commercial (existing, proposed)
  - Delineator (existing, proposed)
  - Portable (proposed)
  - Post mounted (existing, proposed)

UTILITY SYMBOLS

- Irrigation ditch
  - Underground (existing, proposed) — IRRI
  - Surface (existing, proposed) — IRR
- Support pole (existing, proposed)
- Support pole anchor (existing, proposed)
- Street light (existing, proposed)
- Telephone pedestal (existing, proposed) — TP
- Underground utility
  - CATV (existing, proposed) — TVI
  - Fiber optic (existing, proposed) — FOI
  - Gas (existing, proposed) — G
  - Oil (existing, proposed) — O
  - Power (existing, proposed) — P
  - Sanitary sewer (existing, proposed) — SSI
  - Telephone (existing, proposed) — T
  - Water (existing, proposed) — W
- Overhead utility
  - CATV (existing, proposed) — TV
  - Fiber optic (existing, proposed) — FO
  - Power (existing, proposed) — P
  - Telephone (existing, proposed) — T

REGISTERED PROFESSIONAL ENGINEER
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PE-42098
DATE 08/27/2026
NORTH DAKOTA

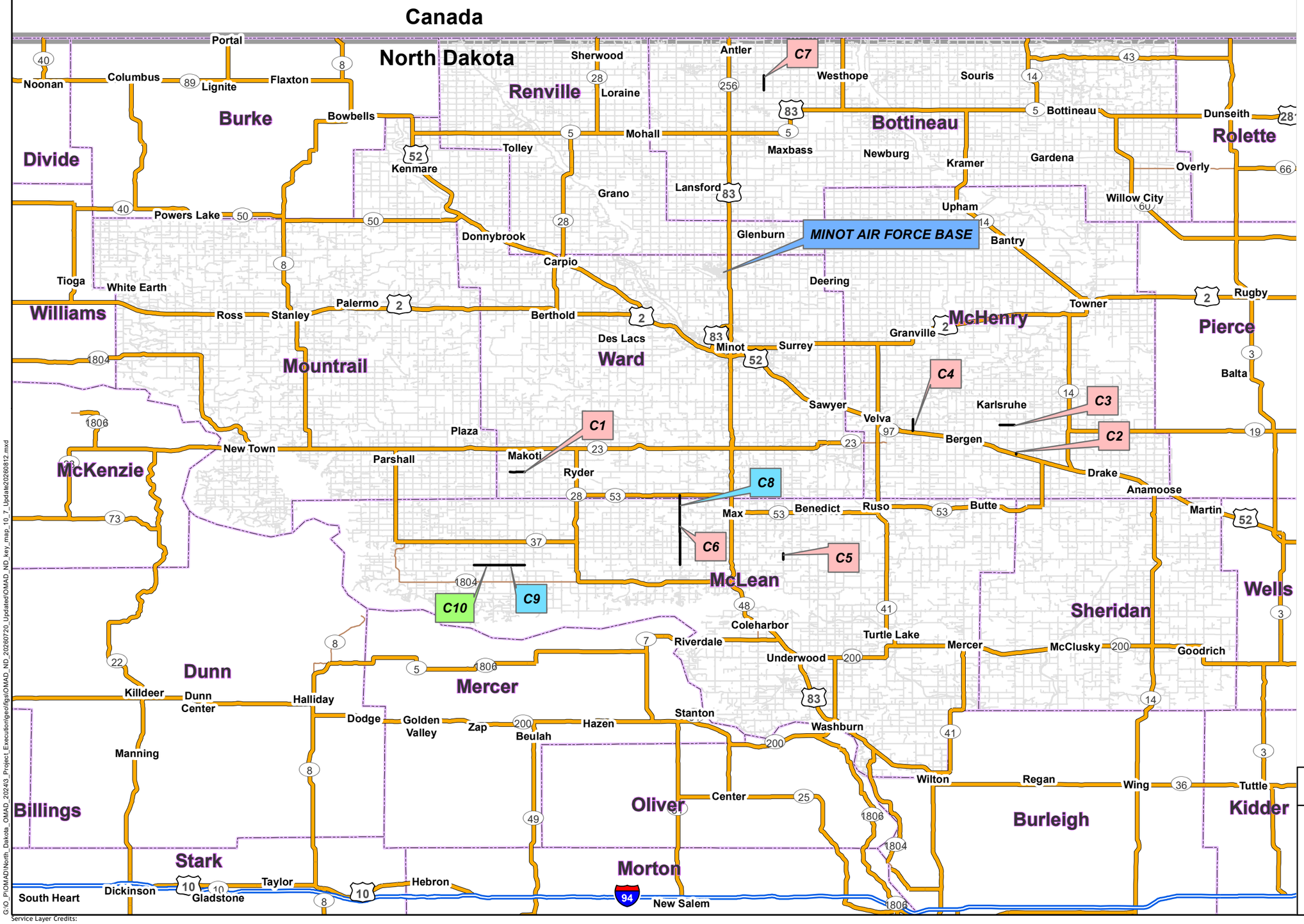
CONVENTIONAL PLAN SYMBOLS AND ABBREVIATIONS

Sheet 2 of 2

...\SYM(MAIN)_ND OMAD 2025(1).dgn [Design IntlFt]

7/21/2026

PROJECT	SHEET NUMBER
ND OMAD 2025(1) MINUTEMAN MISSILE ACCESS ROADS	A4



LEGEND

- C# Schedule A/B/C
- C# Schedule B/C
- C# Schedule C
- County Boundary
- State Boundary

Transportation

- Interstate
- Highways
- Local Road

REGISTERED PROFESSIONAL ENGINEER
CODY A. BURCH
PE-42098
DATE 08/27/2026
NORTH DAKOTA

N

0 5 10 20 Miles

Coordinate System: GCS WGS 1984

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
CENTRAL FEDERAL LANDS HIGHWAY DIVISION

KEY MAP

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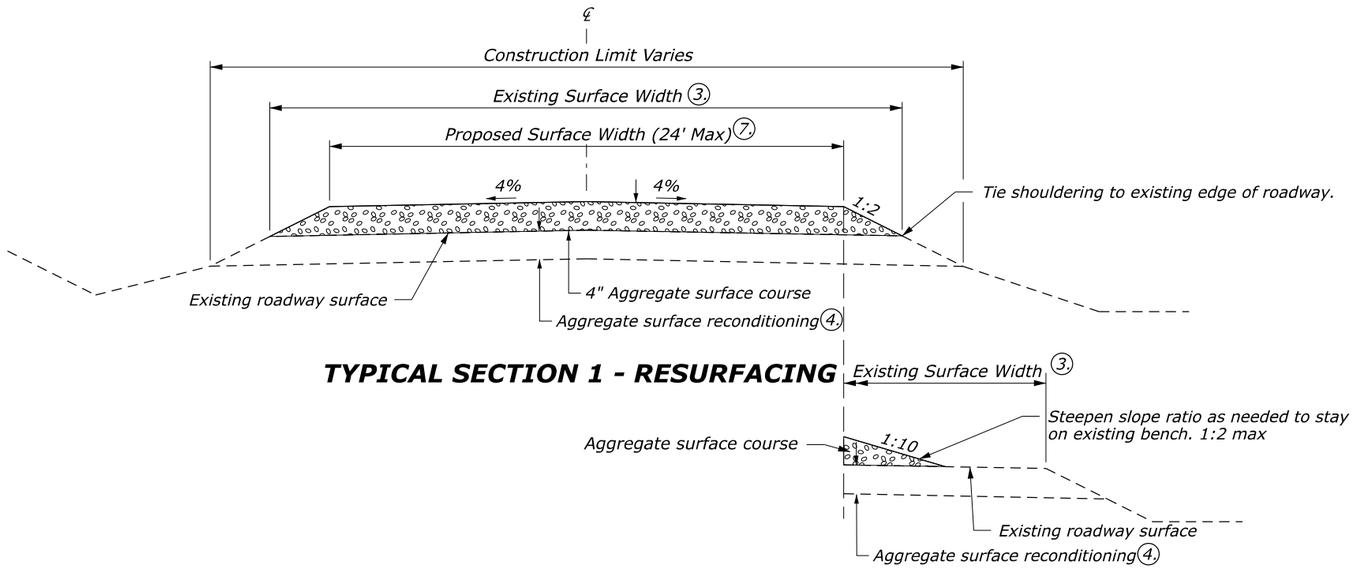
Service Layer Credits:

PROJECT	SHEET NUMBER
ND OMAD 2025(1) MINUTEMAN MISSILE ACCESS ROADS	A5

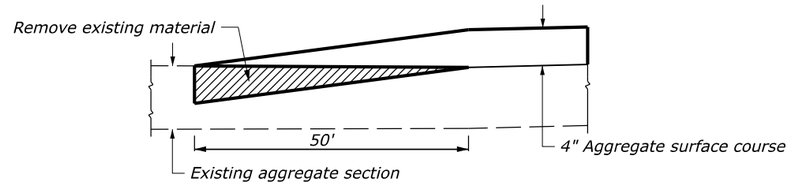


TYPICAL SECTION 1 - RESURFACING

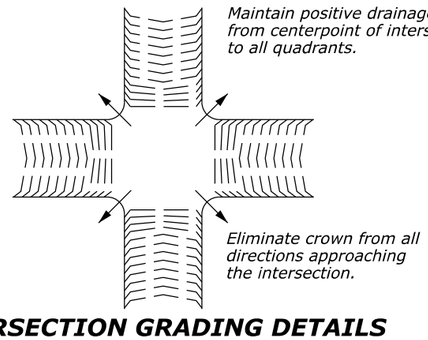
SHOULDERING ALTERNATIVE

NOTE:

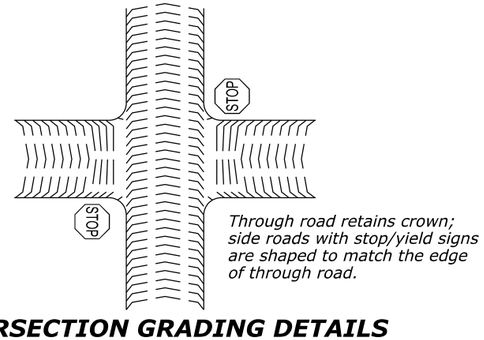
1. Extend resurfacing and reconditioning a minimum of 150 feet beyond the missile silo's approach when road continues.
2. Eliminate 4% crown within intersection at uncontrolled public roadway intersections unless approved to remain. See Intersection Details on this sheet. Begin transition 100 feet from intersection on each intersection approach.
3. See Sheet B5 for existing roadway widths.
4. Perform aggregate surface reconditioning per Section 303. Meet the crown requirements as shown in Typical Section 1 or as approved.
5. Adjust field entrances and approach roads identified during construction to tie to the proposed mainline construction. Match the existing vertical grade. Do not exceed ±10% grade or as approved. Match mainline structural section depth and existing field entrance or approach road width. Locations are not shown in the plans. Work is considered incidental to mainline construction.
6. Transition proposed top of grade to match existing where proposed surfacing terminates. See Detail A. Work is incidental to Aggregate Surface Reconditioning.
7. Provide the maximum surfacing width feasible using the proposed aggregate depth and side slope ratios shown on this sheet. Adjust the proposed width to stay on the existing bench.



DETAIL A - PROPOSED SURFACE TRANSITION (6.)



INTERSECTION GRADING DETAILS UNCONTROLLED INTERSECTION (2.)



INTERSECTION GRADING DETAILS CONTROLLED INTERSECTION (2.)

REGISTERED PROFESSIONAL ENGINEER
CODY A. BURCH
PE-42098
DATE 08/27/2026
NORTH DAKOTA

TYPICAL SECTION

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9/2/2026

SUMMARY OF QUANTITIES - Schedule A

PROJECT	SHEET NUMBER
ND OMAD 2025(1) MINUTEMAN MISSILE ACCESS ROADS	B1

AMEND	LINE ITEM NO.	PAY ITEM NUMBER	PAY ITEM DESCRIPTION	UNIT	Sheet and Description		ALLOWANCE	Estimated Quantities	Remarks and/or Determination of Estimated Quantity
					B5	B6			
					SURFACING SUMMARY	DRAINAGE SUMMARY		BID SCHEDULE	
	A0020	15101-0000	MOBILIZATION	LPSM				ALL	
	A0040	15215-3000	SURVEY AND STAKING, DRAINAGE STRUCTURE	EACH		3		3	
	A0060	15301-0000	CONTRACTOR QUALITY CONTROL	LPSM				ALL	
	A0080	15401-0000	CONTRACTOR TESTING	LPSM				ALL	
	A0100	15501-0000	CONSTRUCTION SCHEDULE	LPSM				ALL	
	A0120	15701-0000	SOIL EROSION CONTROL	LPSM				ALL	
	A0140	15802-0000	WATERING FOR DUST CONTROL	LPSM				ALL	
	A0160	20301-1900	REMOVAL OF PIPE CULVERT	EACH		3		3	
	A0180	20402-0000	SUBEXCAVATION	CUYD				50	As approved
	A0200	20420-0000	EMBANKMENT CONSTRUCTION	CUYD		6		6	
	A0220	30110-0000	AGGREGATE SURFACE COURSE	TON	24520		980	25500	
	A0240	30110-0000	AGGREGATE SURFACE COURSE,(HAUL ROADS)	TON				500	As approved
	A0260	30301-5000	AGGREGATE SURFACE RECONDITIONING	MILE	9.45			9.45	
	A0280	60201-0800	24-INCH PIPE CULVERT,(CMP)	LNFT		35	5	40	
	A0300	60201-0800	24-INCH PIPE CULVERT,(CONCRETE)	LNFT		118	7	125	
	A0320	60210-0800	END SECTION FOR 24-INCH PIPE CULVERT,(CMP)	EACH		2		2	
	A0340	60210-0800	END SECTION FOR 24-INCH PIPE CULVERT,(CONCRETE)	EACH		4		4	
	A0360	62201-0200	DUMP TRUCK, 8 CUBIC YARD MINIMUM CAPACITY	HOUR				10	
	A0380	62201-0950	WHEEL LOADER, 3 CUBIC YARD MINIMUM RATED CAPACITY	HOUR				10	
	A0400	62201-2850	MOTOR GRADER, 12 FOOT MINIMUM BLADE	HOUR				20	
	A0420	62201-3150	HYDRAULIC EXCAVATOR, 1.0 CUBIC YARD MINIMUM CAPACITY WITH THUMB ATTACHMENT	HOUR				10	
	A0440	62301-0000	GENERAL LABOR	HOUR				15	
	A0460	62302-1100	SPECIAL LABOR, HIRED SURVEY SERVICES	HOUR				20	
	A0480	63308-2000	OBJECT MARKER, TYPE 2,(OM2-2V)	EACH		6		6	
	A0500	63501-0000	TEMPORARY TRAFFIC CONTROL	LPSM				ALL	

Milestone: 100%
Date Completed: 9/3/2026
Report date: 9/3/2026

REGISTERED PROFESSIONAL ENGINEER
CODY A. BURCH
PE-42098
DATE 08/27/2026
NORTH DAKOTA

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SUMMARY OF QUANTITIES - Schedule B

PROJECT	SHEET NUMBER
ND OMAD 2025(1) MINUTEMAN MISSILE ACCESS ROADS	B2

AMEND	LINE ITEM NO.	PAY ITEM NUMBER	PAY ITEM DESCRIPTION	UNIT	Sheet and Description		ALLOWANCE	Estimated Quantities	Remarks and/or Determination of Estimated Quantity
					B5	B6			
					SURFACING SUMMARY	DRAINAGE SUMMARY		BID SCHEDULE	
	B0020	15101-0000	MOBILIZATION	LPSM				ALL	
	B0040	15215-3000	SURVEY AND STAKING, DRAINAGE STRUCTURE	EACH		6		6	
	B0060	15301-0000	CONTRACTOR QUALITY CONTROL	LPSM				ALL	
	B0080	15401-0000	CONTRACTOR TESTING	LPSM				ALL	
	B0100	15501-0000	CONSTRUCTION SCHEDULE	LPSM				ALL	
	B0120	15701-0000	SOIL EROSION CONTROL	LPSM				ALL	
	B0140	15802-0000	WATERING FOR DUST CONTROL	LPSM				ALL	
	B0160	20301-1900	REMOVAL OF PIPE CULVERT	EACH		6		6	
	B0180	20402-0000	SUBEXCAVATION	CUYD				75	As approved
	B0200	20420-0000	EMBANKMENT CONSTRUCTION	CUYD		12		12	
	B0220	30110-0000	AGGREGATE SURFACE COURSE	TON	42772		1228	44000	
	B0240	30110-0000	AGGREGATE SURFACE COURSE,(HAUL ROADS)	TON				850	As approved
	B0260	30301-5000	AGGREGATE SURFACE RECONDITIONING	MILE	15.72			15.72	
	B0280	60201-0600	18-INCH PIPE CULVERT,(CMP)	LNFT		38	7	45	
	B0300	60201-0800	24-INCH PIPE CULVERT,(CMP)	LNFT		35	5	40	
	B0320	60201-0800	24-INCH PIPE CULVERT,(CONCRETE)	LNFT		249	11	260	
	B0340	60210-0600	END SECTION FOR 18-INCH PIPE CULVERT,(CMP)	EACH		2		2	
	B0360	60210-0800	END SECTION FOR 24-INCH PIPE CULVERT,(CMP)	EACH		2		2	
	B0380	60210-0800	END SECTION FOR 24-INCH PIPE CULVERT,(CONCRETE)	EACH		8		8	
	B0400	62201-0200	DUMP TRUCK, 8 CUBIC YARD MINIMUM CAPACITY	HOUR				15	
	B0420	62201-0950	WHEEL LOADER, 3 CUBIC YARD MINIMUM RATED CAPACITY	HOUR				15	
	B0440	62201-2850	MOTOR GRADER, 12 FOOT MINIMUM BLADE	HOUR				30	
	B0460	62201-3150	HYDRAULIC EXCAVATOR, 1.0 CUBIC YARD MINIMUM CAPACITY WITH THUMB ATTACHMENT	HOUR				15	
	B0480	62301-0000	GENERAL LABOR	HOUR				20	
	B0500	62302-1100	SPECIAL LABOR, HIRED SURVEY SERVICES	HOUR				30	
	B0520	63308-2000	OBJECT MARKER, TYPE 2,(OM2-2V)	EACH		12		12	
	B0540	63501-0000	TEMPORARY TRAFFIC CONTROL	LPSM				ALL	

Milestone: 100%
Date Completed: 9/3/2026
Report date: 9/3/2026

REGISTERED PROFESSIONAL ENGINEER
CODY A. BURCH
PE-42098
DATE 08/27/2026
NORTH DAKOTA

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SUMMARY OF QUANTITIES - Schedule C

PROJECT	SHEET NUMBER
ND OMAD 2025(1) MINUTEMAN MISSILE ACCESS ROADS	B3

AMEND	LINE ITEM NO.	PAY ITEM NUMBER	PAY ITEM DESCRIPTION	UNIT	Sheet and Description		ALLOWANCE	Estimated Quantities	Remarks and/or Determination of Estimated Quantity
					B5	B6			
					SURFACING SUMMARY	DRAINAGE SUMMARY		BID SCHEDULE	
	C0020	15101-0000	MOBILIZATION	LPSM				ALL	
	C0040	15215-3000	SURVEY AND STAKING, DRAINAGE STRUCTURE	EACH		7		7	
	C0060	15301-0000	CONTRACTOR QUALITY CONTROL	LPSM				ALL	
	C0080	15401-0000	CONTRACTOR TESTING	LPSM				ALL	
	C0100	15501-0000	CONSTRUCTION SCHEDULE	LPSM				ALL	
	C0120	15701-0000	SOIL EROSION CONTROL	LPSM				ALL	
	C0140	15802-0000	WATERING FOR DUST CONTROL	LPSM				ALL	
	C0160	20301-1900	REMOVAL OF PIPE CULVERT	EACH		7		7	
	C0180	20402-0000	SUBEXCAVATION	CUYD				85	As approved
	C0200	20420-0000	EMBANKMENT CONSTRUCTION	CUYD		14		14	
	C0220	30110-0000	AGGREGATE SURFACE COURSE	TON	55535		1665	57200	
	C0240	30110-0000	AGGREGATE SURFACE COURSE,(HAUL ROADS)	TON				1000	As approved
	C0260	30301-5000	AGGREGATE SURFACE RECONDITIONING	MILE	20.27			20.27	
	C0280	60201-0600	18-INCH PIPE CULVERT,(CMP)	LNFT		76	4	80	
	C0300	60201-0800	24-INCH PIPE CULVERT,(CMP)	LNFT		35	5	40	
	C0320	60201-0800	24-INCH PIPE CULVERT,(CONCRETE)	LNFT		249	11	260	
	C0340	60210-0600	END SECTION FOR 18-INCH PIPE CULVERT,(CMP)	EACH		4		4	
	C0360	60210-0800	END SECTION FOR 24-INCH PIPE CULVERT,(CMP)	EACH		2		2	
	C0380	60210-0800	END SECTION FOR 24-INCH PIPE CULVERT,(CONCRETE)	EACH		8		8	
	C0400	60704-0000	CLEANING CULVERT IN PLACE	EACH		1		1	
	C0420	62201-0200	DUMP TRUCK, 8 CUBIC YARD MINIMUM CAPACITY	HOUR				20	
	C0440	62201-0950	WHEEL LOADER, 3 CUBIC YARD MINIMUM RATED CAPACITY	HOUR				20	
	C0460	62201-2850	MOTOR GRADER, 12 FOOT MINIMUM BLADE	HOUR				35	
	C0480	62201-3150	HYDRAULIC EXCAVATOR, 1.0 CUBIC YARD MINIMUM CAPACITY WITH THUMB ATTACHMENT	HOUR				20	
	C0500	62301-0000	GENERAL LABOR	HOUR				25	
	C0520	62302-1100	SPECIAL LABOR, HIRED SURVEY SERVICES	HOUR				35	
	C0540	63308-2000	OBJECT MARKER, TYPE 2,(OM2-2V)	EACH		16		16	
	C0560	63501-0000	TEMPORARY TRAFFIC CONTROL	LPSM				ALL	

Milestone: 100%
Date Completed: 9/3/2026
Report date: 9/3/2026

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PE-42098
DATE 08/27/2026
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3-Sep-2026 3:16 PM

Route Improvement Summary

Schedule	Priority No.	Sheet No.	Route Segment ID	State	County	Road No.	Segment Length (Miles)	Proposed Work	Surfacing (See Sheet B5)	Drainage improvements and Signage (See Sheet B6)	REMARKS
SCHEDULE A											
A/B/C	1	C1	MIAB-F-09	ND	Ward	303rd Ave SW	1.72	Surfacing	X		
A/B/C	2	C2	MIAB-A-01	ND	McHenry	Center Ave N	0.24	Surfacing	X		
A/B/C	3	C3	MIAB-A-11	ND	McHenry	43rd St N	1.71	Surfacing	X		
A/B/C	4	C4	MIAB-B-02S	ND	McHenry	12th Ave N	2.00	Surfacing	X		
A/B/C	5	C5	MIAB-C-09N(b)	ND	McLean	27th Ave NW	1.08	Surfacing	X		
A/B/C	6	C6	MIAB-D-08	ND	McLean	39th Ave NW	N/A	Drainage improvements, Signage		X	
A/B/C	7	C7	MIAB-O-03	ND	Bottineau	23rd Ave NW	2.56	Drainage improvements, Surfacing, Signage	X	X	
SCHEDULE A TOTALS							**9.31**				
SCHEDULE B											
B/C	8	C8	MIAB-D-09	ND	McLean/Ward	39th Ave NW/ 86th St SW	4.80	Drainage improvements, Surfacing, Signage	X	X	
B/C	9	C9	MIAB-E-08(b)	ND	McLean	19th St NW	1.41	Drainage improvements, Surfacing, Signage	X	X	
SCHEDULE B TOTALS							**15.52**				
SCHEDULE C											
C	10	C10	MIAB-E-09	ND	McLean	19th St NW	4.49	Drainage improvements, Surfacing, Signage	X	X	
SCHEDULE C TOTALS							**20.01**				

NOTE:

1. Segment lengths are shown for routes containing surfacing improvements. Routes containing only drainage improvements have a segment length shown as "N/A".

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IMPROVEMENT SUMMARY

...\TABS(MAIN)_ND OMAD 2025(1).dgn [Design SurvFt]

9/2/2026

ROADWAY SURFACING SUMMARY

Schedule	Priority No.	Sheet No.	Route ID	Road	Begin	End	Segment Length (See Notes 1 & 2)	Average Existing Surface Width (See Note 3)	30110-0000 AGGREGATE SURFACE COURSE	30301-5000 AGGREGATE SURFACE RECONDITIONING	REMARKS
							MILE	FEET	TON	MILE	
SCHEDULE A											
A/B/C	1	C1	MIAB-F-09	303rd Ave SW	F-09	338th St SW	1.72	21.4	4,513	1.75	End work at gravel/asphalt transition.
A/B/C	2	C2	MIAB-A-01	Center Ave N	HWY 52	A-01	0.24	22.8	680	0.27	Begin work at gravel/asphalt transition.
A/B/C	3	C3	MIAB-A-11	43rd St N	2nd Ave N	A-11	1.71	18.0	3,776	1.74	Begin work at asphalt/gravel transition.
A/B/C	4	C4	MIAB-B-02S	12th Ave N	42nd St N	44th St N	2.00	19.8	4,844	2.00	Begin work at asphalt/gravel transition.
A/B/C	5	C5	MIAB-C-09N(b)	27th Ave NW	C-09	21st St NW	1.08	24'+	3,181	1.11	Begin work at asphalt/gravel transition.
A/B/C	7	C7	MIAB-O-03	23rd Ave NW	O-03	103rd St NW	2.56	24'+	7,526	2.59	End work at gravel/asphalt transition.
SCHEDULE A TOTALS									**24,520**	**9.45**	
				SCHEDULE B							
B/C	8	C8	MIAB-D-09	39th Ave NW/ 86th St SW	D-09	359th Ave SW	4.80	24'+	14,102	4.83	End work at gravel/asphalt transition.
B/C	9	C9	MIAB-E-08(b)	19th St NW	E-08	57th Ave NW	1.41	24'+	4,150	1.44	End work at gravel/asphalt transition.
SCHEDULE B TOTALS									**42,772**	**15.72**	
SCHEDULE C											
C	10	C10	MIAB-E-09	19th St NW	E-09	E-08	4.49	23.2	12,763	4.55	Perform no work for approximately 85 feet on asphalt pavement at the 59th Ave NW intersection.
SCHEDULE C TOTALS									**55,535**	**20.27**	

VALUES USED FOR ESTIMATING PURPOSES
Aggregate Surface Course (Dry unit weight)----------139.0 lb/ft3

NOTE:
1. Per Note 1 on Sheet A5, extend resurfacing and reconditioning a minimum of 150 feet beyond the missile site's appraoch when road continues. Length is not included in the 'Segment Length' shown in the plans. Length included in Aggregate Surface Course and Aggregate Surfacing Reconditioning.
2. Segment Length shown is an approximate length measured along the route centerline and is to be used for information only. Perform required work to the begin and end locations as shown in this table and the C-sheets.
3. Existing surface widths are shown for information only. Proposed aggregate width is 24-feet maximum.
4. Shouldering wedges are included in aggregate quantities. See typical section on Sheet A5 for more information.

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SURFACING SUMMARY

...\TABS(MAIN)_ND OMAD 2025(1).dgn [Design SurvFt]
9/3/2026

PROJECT	SHEET NUMBER
ND OMAD 2025(1) MINUTEMAN MISSILE ACCESS ROADS	B6

DRAINAGE SUMMARY

Schedule	Priority No.	Sheet No.	Route Segment ID	Culvert ID	Segment Milepost	County	Existing Culvert (See Note 1 and 2)			15215-3000 SURVEY AND STAKING, DRAINAGE STRUCTURE	20301-1900 REMOVAL OF PIPE CULVERT	20420-0000 EMBANKMENT CONSTRUCTION (See Note 8)	60201-0600 18-INCH PIPE CULVERT (CMP)	60201-0800 24-INCH PIPE CULVERT (CMP)	60201-0800 24-INCH PIPE CULVERT (CONCRETE)	60210-0600 END SECTION FOR 18-INCH PIPE CULVERT (CMP)	60210-0800 END SECTION FOR 24-INCH PIPE CULVERT (CMP)	60210-0800 END SECTION FOR 24-INCH PIPE CULVERT (CONCRETE)	60704-0000 CLEANING CULVERT IN PLACE	63308-2000 OBJECT MARKER, TYPE 2 (OM2-2V)
							Type	Diameter	Length	EACH	EACH	CUYD	LNFT	LNFT	LNFT	EACH	EACH	EACH	EACH	EACH
SCHEDULE A																				
A/B/C	6	C6	MIAB-D-08	C-0007	2.77	McLean	Concrete	15"	70'	1	1	2			70			2		2
A/B/C	6	C6	MIAB-D-08	C-0016	0.79	McLean	Concrete	18"	51'	1	1	2			48			2		2
A/B/C	7	C7	MIAB-O-03	C-0002	0.32	Bottineau	CMP	24"	38'	1	1	2		35			2			2
SCHEDULE A TOTALS										**3**	**3**	**6**		**35**	**118**		**2**	**4**		**6**
SCHEDULE B																				
B/C	8	C8	MIAB-D-09	C-0004	3.20	McLean/Ward	Concrete	Assumed 18"	42'	1	1	2			41			2		2
B/C	8	C8	MIAB-D-09	C-0006	2.66	McLean/Ward	Concrete	18	94'	1	1	2			90			2		2
B/C	9	C9	MIAB-E-08(b)	C-0002	0.62	McLean	CMP	18"	41'	1	1	2	38			2				2
SCHEDULE B TOTALS										**6**	**6**	**12**	**38**	**35**	**249**	**2**	**2**	**8**		**12**
SCHEDULE C																				
C	10	C10	MIAB-E-09	C-0011	1.20	McLean	CMP	Assumed 18"	41'	1	1	2	38			2				2
C	10	C10	MIAB-E-09	C-0013	0.75	McLean	Concrete	18"	35'										1	2
SCHEDULE C TOTALS										**7**	**7**	**14**	**76**	**35**	**249**	**4**	**2**	**8**	**1**	**16**

NOTE:

1. Existing culvert informtion is for information only. Survey was not completed as part of this project.
2. Assumed dimensions shown for culverts that could not be field verified.
3. Clearing and grubbing is included in the cost of the culvert work and not measured for payment. Clear the minimal amount necessary for culvert maintenance work.
4. Mileposts are measured from route begin locations as shown on C-sheets.
5. Object Marker installation includes the cost of removing conflicting signage. See Special 633-A.
6. Adjust culvert flowline elevations, as approved, to maintain adequate pipe cover material.
7. 15215-3000 Survey and Staking Drinage Structure is only shown for culverts requiring pipe plots in accordance with Section 152.
8. Embankment Construction quantity shown for use to backfill and grade eroded existing ground areas outside of the culvert limits. Use as approved to create positive drainage at the culvert inlet and outlet.

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DRAINAGE SUMMARY

...\TABS(MAIN)_ND OMAD 2025(1).dgn [Design SurvFit]

8/26/2026

PROJECT	SHEET NUMBER
ND OMAD 2025(1) MINUTEMAN MISSILE ACCESS ROADS	C1

0 500 1,000 2,000 Feet

1.72 Miles
MIAB-F-09

338th St SW

303rd Ave SW

352nd St SW

F-09

Begin Typical Section 1 - Surfacing
Schedule A/B/C
Ward County, ND
N = ± 338446.46
E = ± 1638775.09

End Typical Section 1 - Surfacing
Schedule A/B/C
Ward County, ND
N = ± 338303.23
E = ± 1647826.67

LEGEND

Proposed Route

- Typical Section 1 - Surfacing

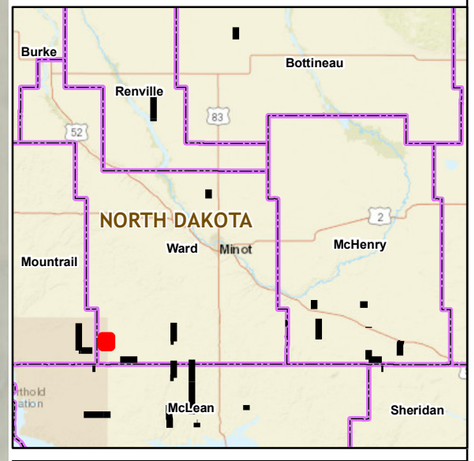
Transportation

- Interstate
- Highways
- Local Road

- County Boundary
- Missile Silo
- Project Culvert

Notes:
Begin/End Northing and Easting - ND State Plane NAD 83.

End work at gravel/asphalt transition.



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ROUTE ID
MIAB-F-09
(SCHEDULE A/B/C)

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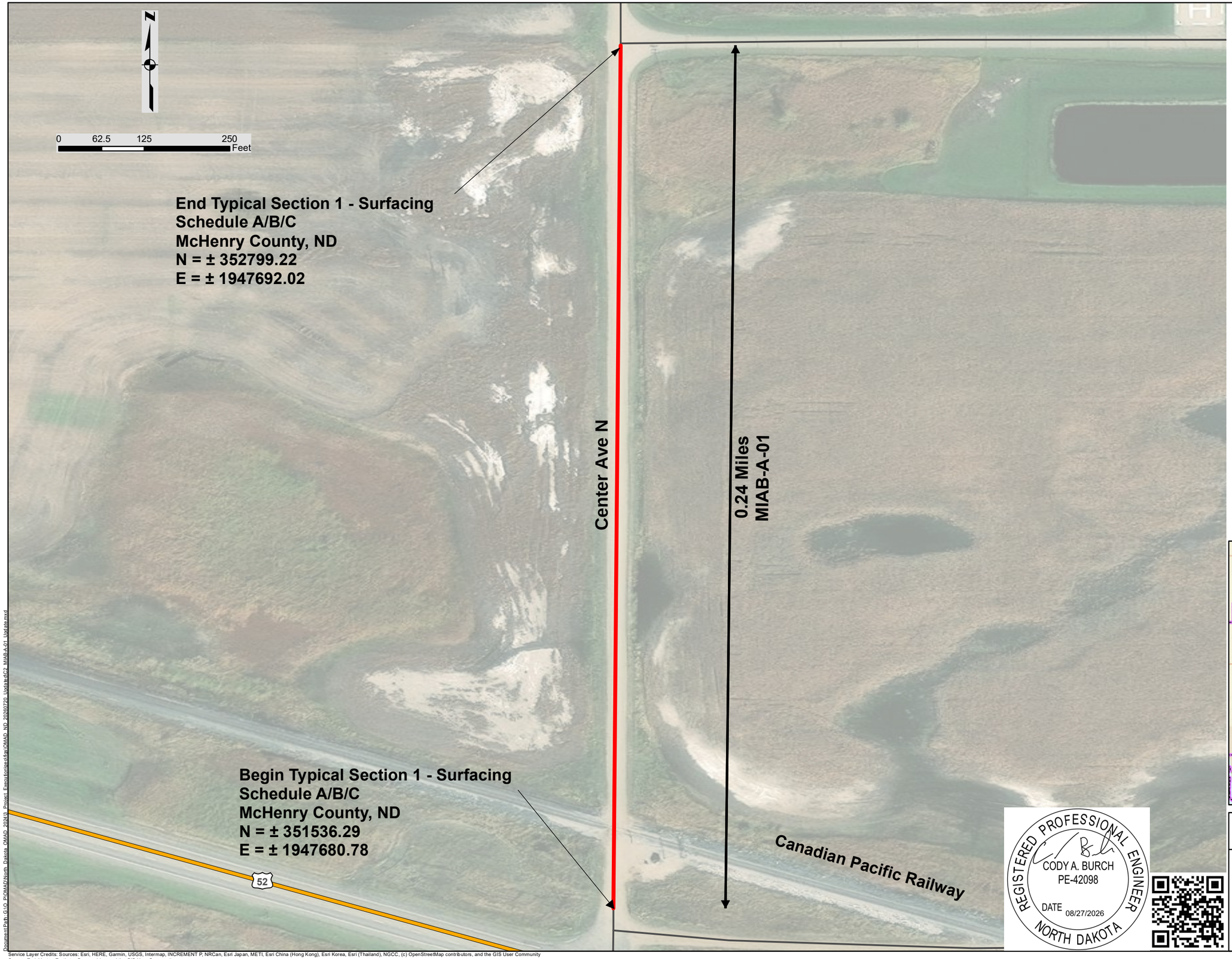
PROJECT	SHEET NUMBER
ND OMAD 2025(1) MINUTEMAN MISSILE ACCESS ROADS	C2



LEGEND

Proposed Route

- Typical Section 1 - Surfacing

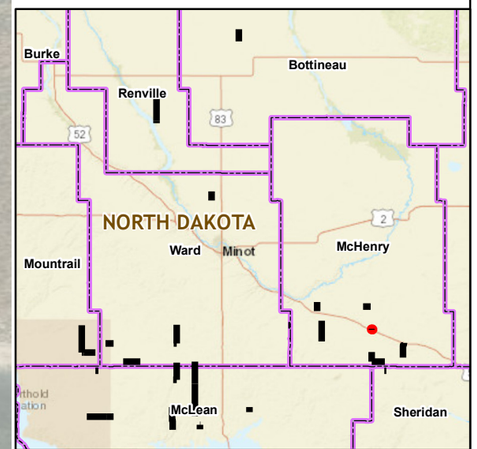
Transportation

- Interstate
- Highways
- Local Road

- County Boundary
- Missile Silo
- Project Culvert

Notes:
Begin/End Northing and Easting - ND State Plane NAD 83.

Begin work at asphalt/gravel transition.



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ROUTE ID
MIAB-A-01
(SCHEDULE A/B/C)

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Source: Esri, Vantor, Earthstar Geographics, and the GIS User Community

PROJECT	SHEET NUMBER
ND OMAD 2025(1) MINUTEMAN MISSILE ACCESS ROADS	C3

N

0 500 1,000 2,000 Feet

2nd Ave N

1.71 Miles
MIAB-A-11

43rd St N

1st Ave N

A-11

Begin Typical Section 1 - Surfacing
Schedule A/B/C
McHenry County, ND
N = ± 378222.87
E = ± 1937698.52

End Typical Section 1 - Surfacing
Schedule A/B/C
McHenry County, ND
N = ± 378205.73
E = ± 1946736.34

LEGEND

Proposed Route

- Typical Section 1 - Surfacing

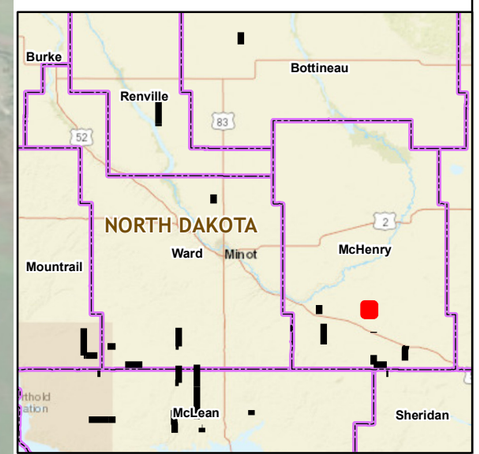
Transportation

- Interstate
- Highways
- Local Road

- County Boundary
- Missile Silo
- Project Culvert

Notes:
Begin/End Northing and Easting - ND State Plane NAD 83.

Begin work at asphalt/gravel transition.



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ROUTE ID
MIAB-A-11
(SCHEDULE A/B/C)

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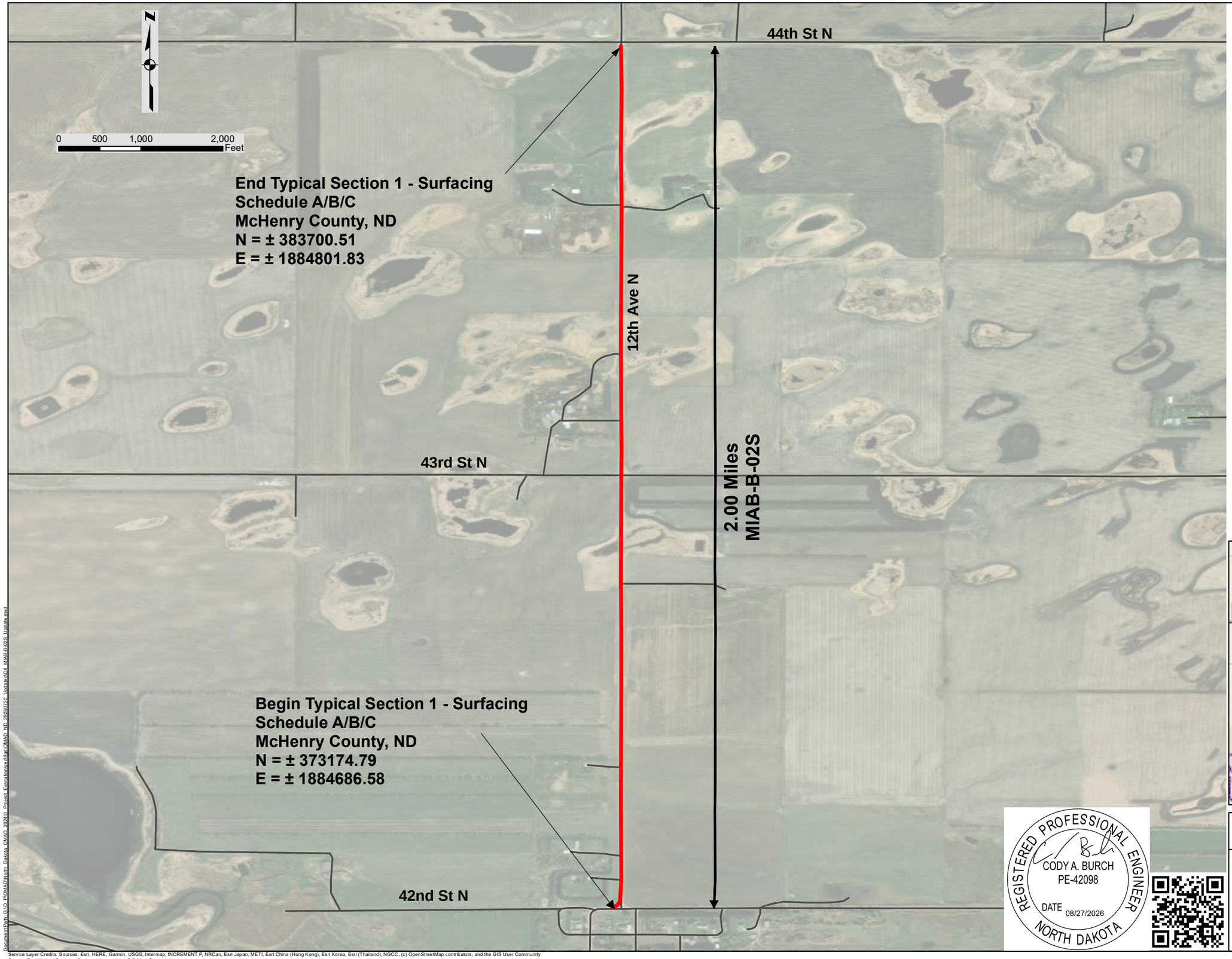
PROJECT	SHEET NUMBER
ND OMAD 2025(1) MINUTEMAN MISSILE ACCESS ROADS	C4



LEGEND

Proposed Route

- Typical Section 1 - Surfacing

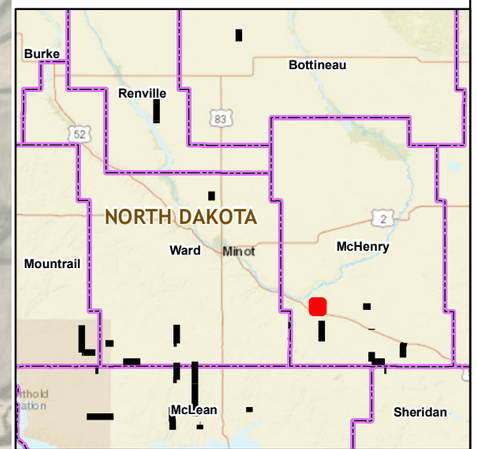
Transportation

- Interstate
- Highways
- Local Road

- County Boundary
- Missile Silo
- Project Culvert

Notes:
Begin/End Northing and Easting - ND State Plane NAD 83.

Begin work at asphalt/gravel transition.



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ROUTE ID
MIAB-B-02S
(SCHEDULE A/B/C)

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Source: Esri, Vantor, Earthstar Geographics, and the GIS User Community

PROJECT	SHEET NUMBER
ND OMAD 2025(1) MINUTEMAN MISSILE ACCESS ROADS	C5

21st St NW

0 285 570 1,140 Feet

End Typical Section 1 - Surfacing
Schedule A/B/C
McLean County, ND
N = ± 262524.42
E = ± 1804925.29

27th Ave NW

1.08 Miles
MIAB-C-09N(b)

Begin Typical Section 1 - Surfacing
Schedule A/B/C
McLean County, ND
N = ± 256839.81
E = ± 1804875.23

20th St NW

C-09

LEGEND

Proposed Route

Typical Section 1 - Surfacing

Transportation

Interstate

Highways

Local Road

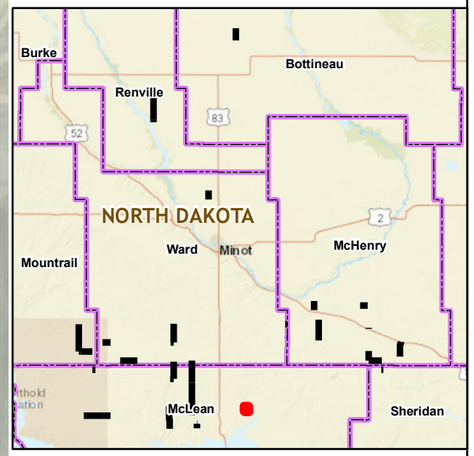
County Boundary

Missile Silo

Project Culvert

Notes:
Begin/End Northing and Easting - ND State Plane NAD 83.

Begin work at asphalt/gravel transition



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DATE 08/27/2026
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ROUTE ID
MIAB-C-09N(b)
(SCHEDULE A/B/C)

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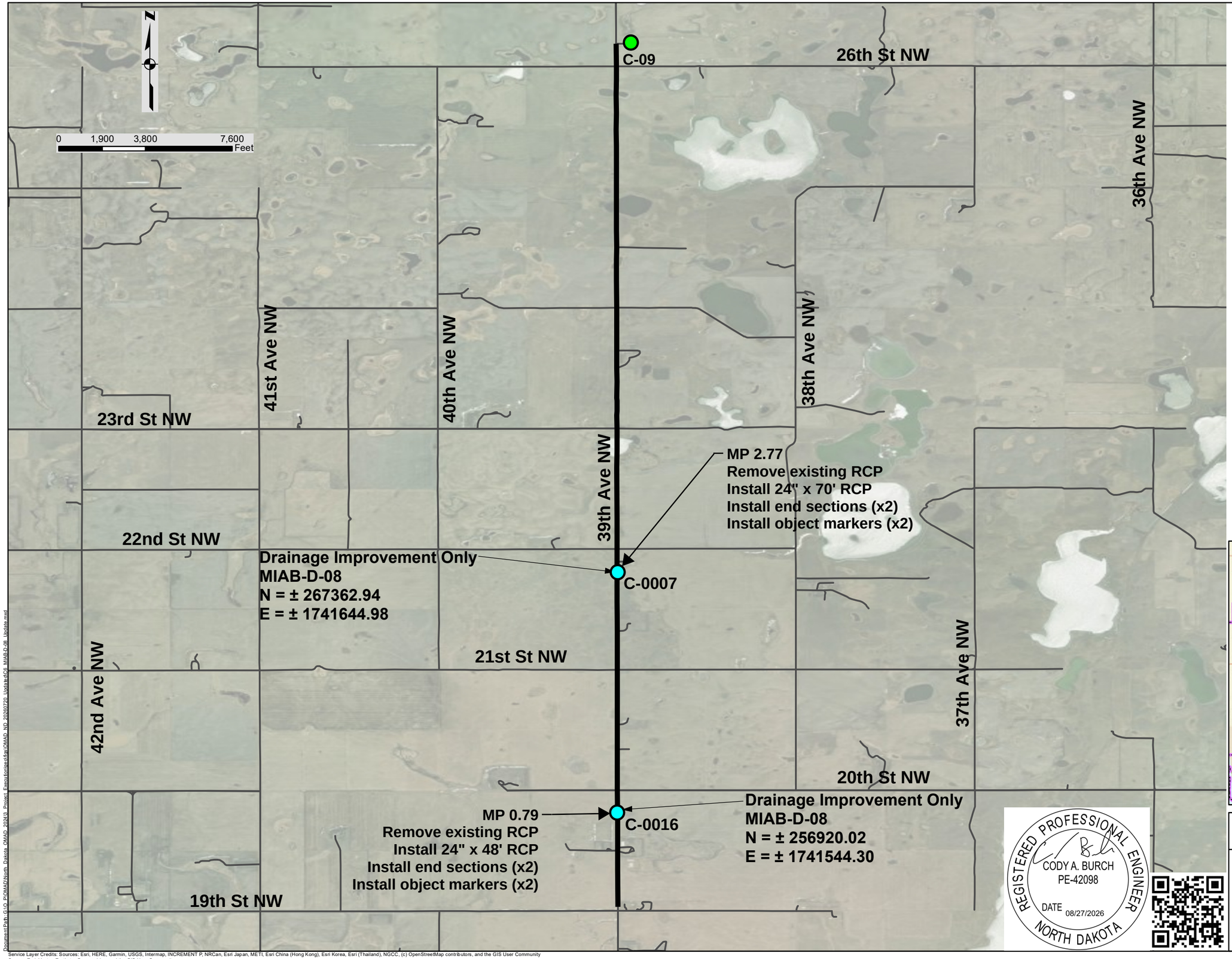
PROJECT	SHEET NUMBER
ND OMAD 2025(1) MINUTEMAN MISSILE ACCESS ROADS	C6



LEGEND

Proposed Route

- *Typical Section 1 - Surfacing*
- *Drainage Improvement Route*

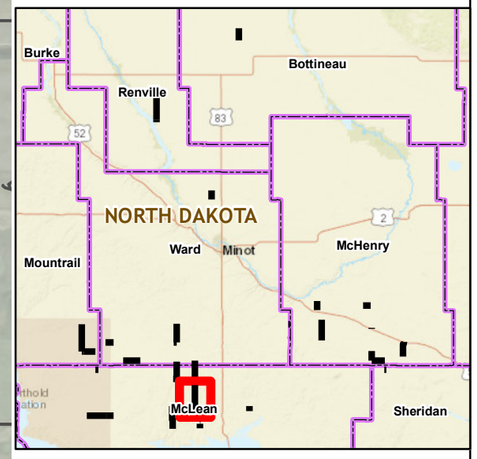
Transportation

- *Interstate*
- *Highways*
- *Local Road*

- *County Boundary*
- *Missile Silo*
- *Project Culvert*

Notes:
Begin/End Northing and Easting - ND State Plane NAD 83.

Begin work at asphalt/gravel transition.



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NORTH DAKOTA

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ROUTE ID
MIAB-D-08
(SCHEDULE A/B/C)

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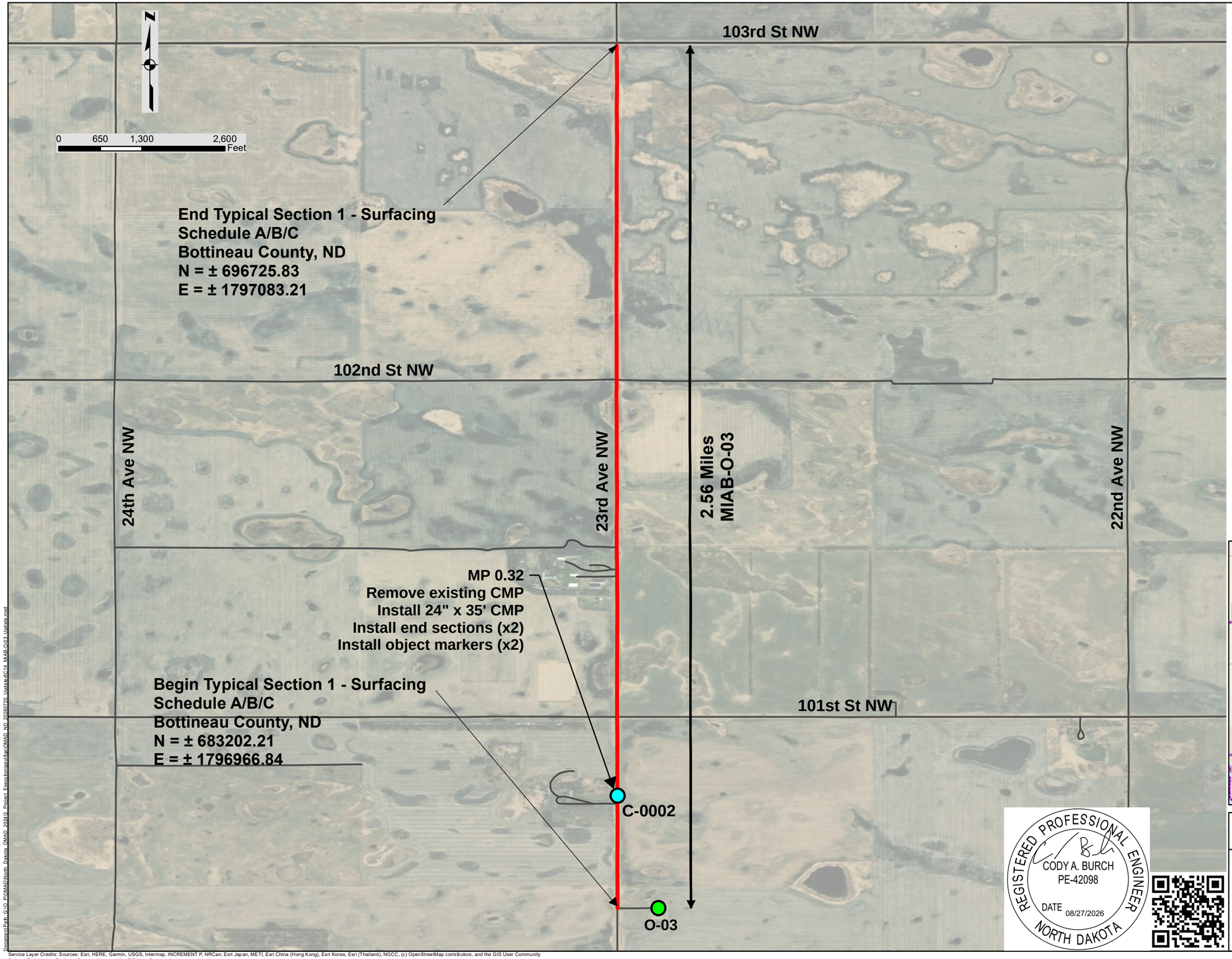
PROJECT	SHEET NUMBER
ND OMAD 2025(1) MINUTEMAN MISSILE ACCESS ROADS	C7



LEGEND

Proposed Route

- Typical Section 1 - Surfacing

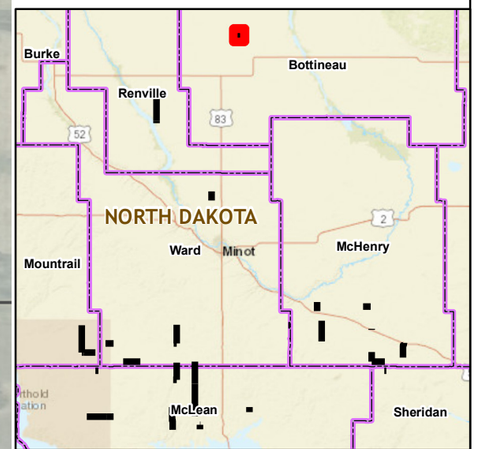
Transportation

- Interstate
- Highways
- Local Road

- County Boundary
- Missile Silo
- Project Culvert

Notes:
Begin/End Northing and Easting - ND State Plane NAD 83.

End work at gravel/asphalt transition.



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CENTRAL FEDERAL LANDS HIGHWAY DIVISION

ROUTE ID
MIAB-O-03
(SCHEDULE A/B/C)

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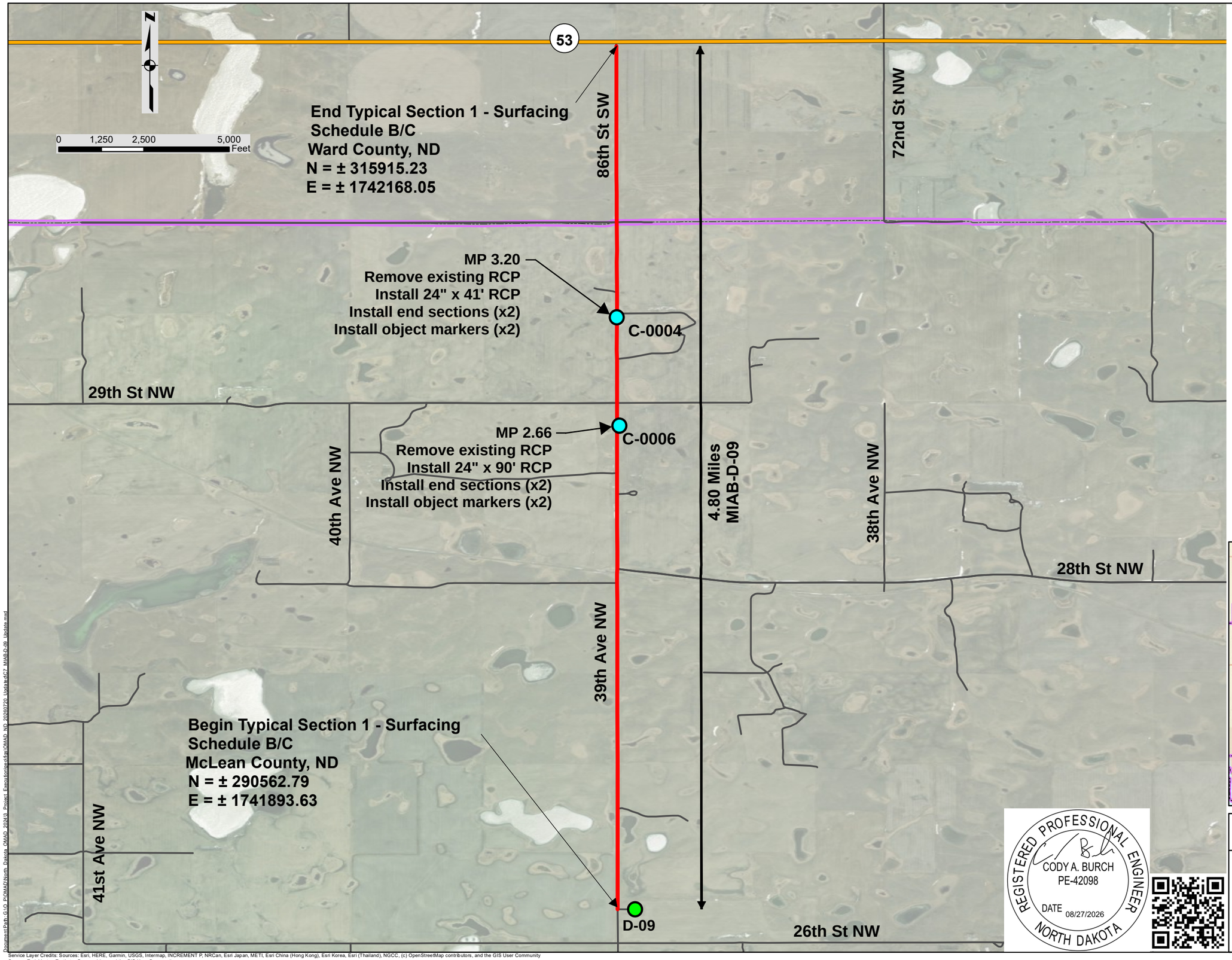
PROJECT	SHEET NUMBER
ND OMAD 2025(1) MINUTEMAN MISSILE ACCESS ROADS	C8



LEGEND

Proposed Route

- Typical Section 1 - Surfacing

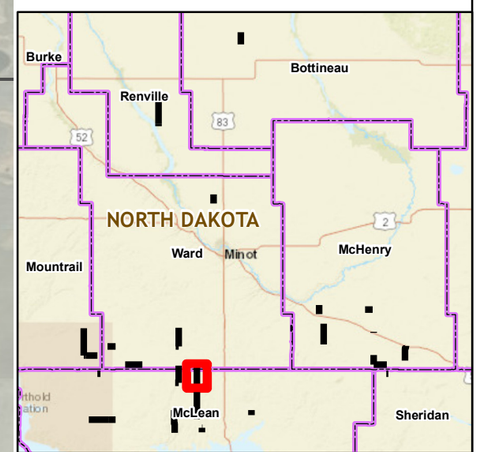
Transportation

- Interstate
- Highways
- Local Road

- County Boundary
- Missile Silo
- Project Culvert

Notes:
Begin/End Northing and Easting - ND State Plane NAD 83.

End work at gravel/asphalt transition.



REGISTERED PROFESSIONAL ENGINEER
CODY A. BURCH
PE-42098
DATE 08/27/2026
NORTH DAKOTA

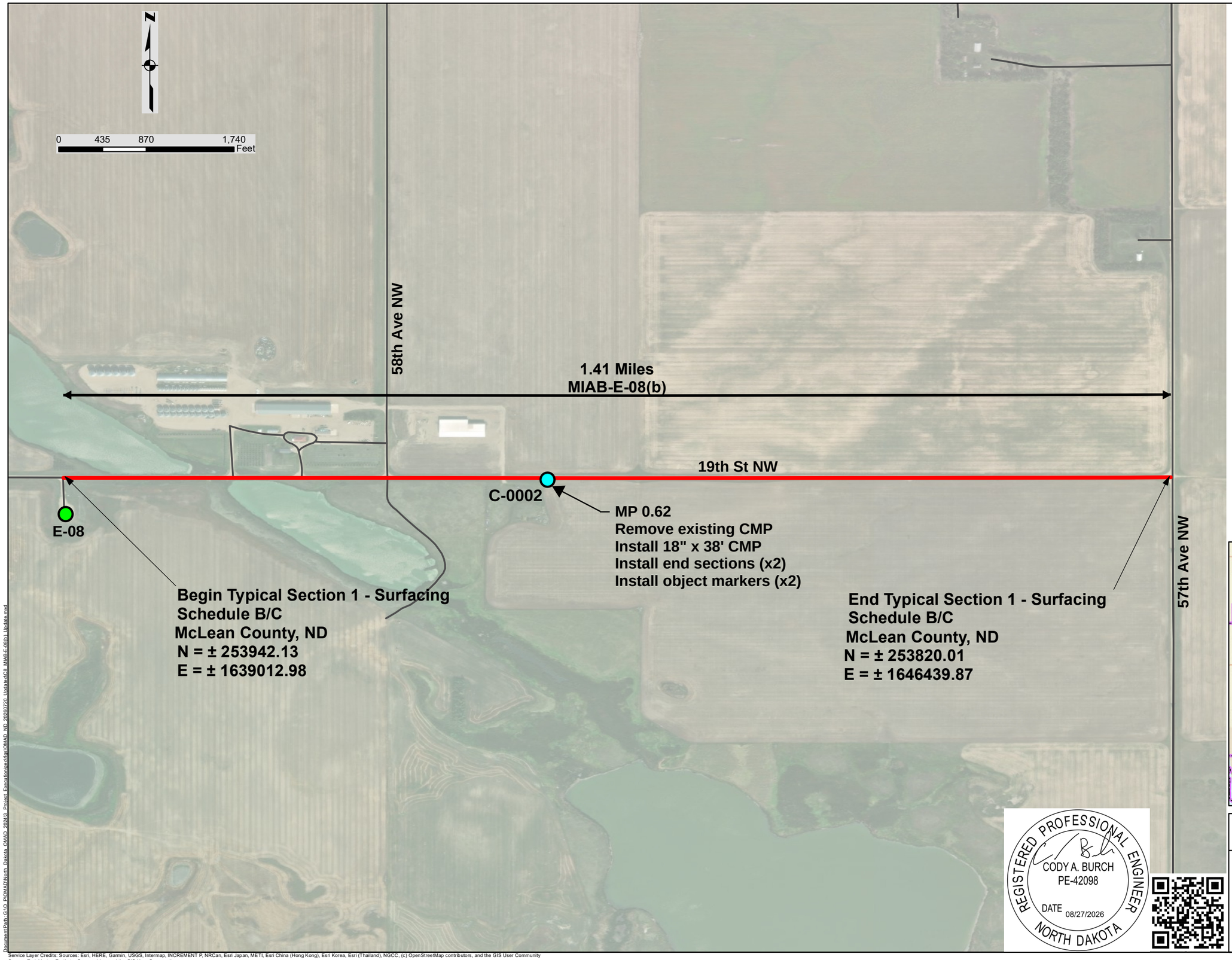
U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
CENTRAL FEDERAL LANDS HIGHWAY DIVISION

ROUTE ID
MIAB-D-09
(SCHEDULE B/C)

DocumentPath: G:\O_P\OMAD\North_Dakota_OMAD_2025\3_Project_Execution\mxd\arc\OMAD_ND_20260720_Update\C7_MIAB-D-09_Update.mxd

Service Layer Credits: Sources: Esri, HERE, Garmin, USGS, Intermap, INCREMENT P, NRCan, Esri Japan, METI, Esri China (Hong Kong), Esri Korea, Esri (Thailand), NGCC, (c) OpenStreetMap contributors, and the GIS User Community
Source: Esri, Vantor, Earthstar Geographics, and the GIS User Community

PROJECT	SHEET NUMBER
ND OMAD 2025(1) MINUTEMAN MISSILE ACCESS ROADS	C9



LEGEND

Proposed Route

Typical Section 1 - Surfacing

Transportation

Interstate

Highways

Local Road

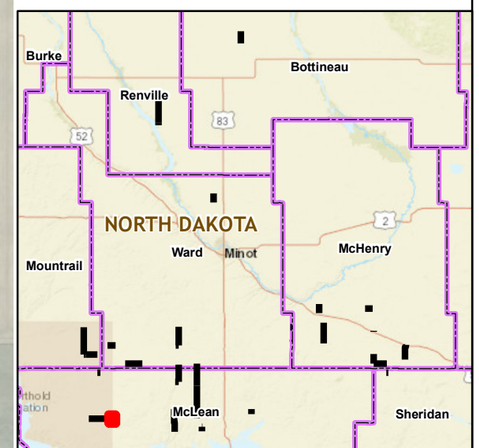
County Boundary

Missile Silo

Project Culvert

Notes:
Begin/End Northing and Easting - ND State Plane NAD 83.

End work at asphalt/gravel transition



REGISTERED PROFESSIONAL ENGINEER
CODY A. BURCH
PE-42098
DATE 08/27/2026
NORTH DAKOTA

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
CENTRAL FEDERAL LANDS HIGHWAY DIVISION

ROUTE ID
MIAB-E-08(b)
(SCHEDULE B/C)

DocumentPath: G:\O_P\OMAD\North_Dakota_OMAD_2024\3_Project_Execution\mxd\sp\OMAD_ND_20260720_Update\C9_MIAB-E-08(b)_Update.mxd

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Source: Esri, Vantor, Earthstar Geographics, and the GIS User Community

PROJECT	SHEET NUMBER
ND OMAD 2025(1) MINUTEMAN MISSILE ACCESS ROADS	C10

0 1,375 2,750 5,500 Feet

61st Ave NW

60th Ave NW

59th Ave NW

20th St NW

MP 0.75
Existing RCP
Cleaning culvert in place
Install object markers (x2)

MP 1.20
Remove existing CMP
Install 18" x 38' CMP
Install end sections (x2)
Install object markers (x2)

4.49 Miles
MIAB-E-09

E-09

C-0013

C-0011

19th St NW

E-08

65th Ave NW

Begin Typical Section 1 - Surfacing
Schedule C
McLean County, ND
N = ± 254383.30
E = ± 1615239.08

End Typical Section 1 - Surfacing
Schedule C
McLean County, ND
N = ± 253942.13
E = ± 1639012.98

18th St NW

65th Ave NW

61st Ave NW

LEGEND

Proposed Route

- Typical Section 1 - Surfacing

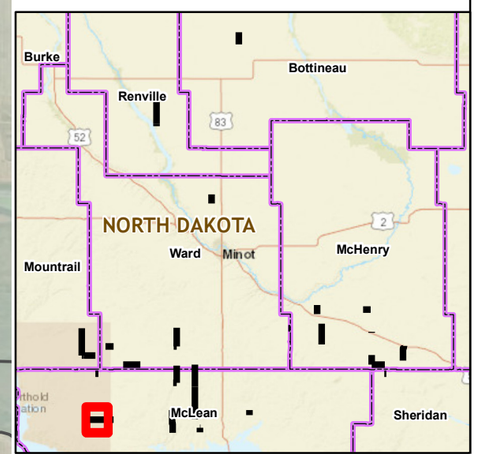
Transportation

- Interstate
- Highways
- Local Road

- County Boundary
- Missile Silo
- Project Culvert

Notes:
Begin/End Northing and Easting - ND State Plane NAD 83.

Perform no work for approximately 85 feet on asphalt pavement at the 59th Ave NW intersection.



REGISTERED PROFESSIONAL ENGINEER
CODY A. BURCH
PE-42098
DATE 08/27/2026
NORTH DAKOTA

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
CENTRAL FEDERAL LANDS HIGHWAY DIVISION

ROUTE ID
MIAB-E-09
(SCHEDULE C)

DocumentPath: G:\O_P\OMAD\North_Dakota_OMAD_2024\3_Project_Execution\mxd\esri\OMAD_ND_20260720_Update\C09_MIABE-09_Update.mxd

Service Layer Credits: Sources: Esri, HERE, Garmin, USGS, Intermap, INCREMENT P, NRCan, Esri Japan, METI, Esri China (Hong Kong), Esri Korea, Esri (Thailand), NGCC, (c) OpenStreetMap contributors, and the GIS User Community
Source: Esri, Vantor, Earthstar Geographics, and the GIS User Community

PROJECT	SHEET NUMBER
ND OMAD 2025(1) MINUTEMAN MISSILE ACCESS ROADS	T1

METAL ROUND PIPE CULVERT

FILL HEIGHT AND METAL THICKNESS TABLE FOR HELICAL LOCKSEAM AND WELDED SEAM PIPE CULVERT

STEEL

PIPE SIZE DIAMETER INCHES	MINIMUM COVER INCHES	2⅔" x ½" CORRUGATIONS					3" x 1" CORRUGATIONS					5" x 1" CORRUGATIONS				
		METAL THICKNESS (INCH/GAGE)														
		0.064/16	0.079/14	0.109/12	0.138/10	0.168/8	0.064/16	0.079/14	0.109/12	0.138/10	0.168/8	0.064/16	0.079/14	0.109/12	0.138/10	0.168/8
		MAXIMUM FILL HEIGHT ABOVE TOP OF PIPE (FEET)														
12	12	100	100	100	100	100										
15	12	100	100	100	100	100										
18	12	100	100	100	100	100										
21	12	100	100	100	100	100										
24	12	100	100	100	100	100										
30	12	85	100	100	100	100										
36	12	71	89	100	100	100	81	100	100	100	100					
42	12	61	76	100	100	100	70	87	100	100	100					
48	12	53	66	93	100	100	61	76	100	100	100	54	68	95	100	100
54	12		59	83	100	100	54	68	95	100	100	48	60	85	100	100
60	12			74	97	100	49	61	86	100	100	43	54	76	98	100
66	12				87	100	44	55	78	100	100	39	49	69	89	100
72	12				80	97	40	51	71	92	100	36	45	63	82	100
78	12					87	37	47	66	85	100	33	42	58	75	92
84	12					75	35	43	61	78	96	31	39	54	70	86
90	12						32	40	57	73	90	29	36	51	65	80
96	12							38	53	69	84		34	48	61	75
102	18							36	50	65	79		32	45	57	71
108	18								47	61	75			42	54	67
114	18								45	58	71			40	52	63
120	18								43	55	67			38	49	60
126	18									52	64				47	57
132	18									50	61				44	54
138	18									48	58				42	52
144	18										56					50

ALUMINUM

PIPE SIZE DIAMETER INCHES	MINIMUM COVER INCHES	2⅔" x ½" CORRUGATIONS					3" x 1" CORRUGATIONS				
		METAL THICKNESS (INCH/GAGE)									
		0.060/16	0.075/14	0.105/12	0.135/10	0.164/8	0.060/16	0.075/14	0.105/12	0.135/10	0.164/8
		MAXIMUM FILL HEIGHT ABOVE TOP OF PIPE (FEET)									
12	12	100	100	100	100	100					
15	12	100	100	100	100	100					
18	12	100	100	100	100	100					
21	12	88	100	100	100	100					
24	12	77	97	100	100	100					
30	12	62	77	100	100	100	71	89	100	100	100
36	12	52	64	90	100	100	59	74	100	100	100
42	12	44	55	77	99	100	51	64	89	100	100
48	12			67	87	100	44	56	78	100	100
54	18			54	71	88	39	50	69	93	100
60	18				57	72	35	45	62	83	98
66	18					58	32	40	56	76	89
72	18					45	30	37	55	70	82
78	24							34	48	64	75
84	24								44	59	70
90	24								41	62	65
96	24								38	51	61
102	24									46	55
108	24									42	50
114	24										45
120	24										40

NOTE:

1. When directed, camber pipe culverts upward from a chord through the inlet and outlet inverts an ordinate amount equal to 1% of the pipe length. Develop camber on a parabolic curve. If the midpoint elevation on the parabolic curve as designed exceeds the elevation of the inlet invert, reduce the amount of camber or increase the pipe culvert gradient.
2. Fill heights exceeding 100 feet require special analysis by the CO.
3. The fill heights in the table are for helical lockseam and welded seam pipe only. Fill heights for culvert pipe with annular corrugations are more restrictive than those of helical lockseam and welded seam pipe. Obtain approval before furnishing annular corrugation pipe.
4. For flexible pavement and aggregate surface roadways, measure minimum cover from the top of the pipe culvert to the bottom of the roadway subgrade. For rigid pavements, measure minimum cover from the top of the pipe culvert to the top of the pavement. For all roadway surface types, measure maximum fill height from the top of the pipe culvert to the top of the pavement.

My responsibility with respect to this plan sheet is limited to the selection of the standard plan for this project and a determination that the selection is appropriate for the project

METAL PIPE ARCH CULVERT

FILL HEIGHT AND METAL THICKNESS TABLE FOR HELICAL LOCKSEAM AND WELDED SEAM PIPE CULVERT

STEEL

PIPE ARCH SIZE SPAN x RISE INCHES	EQUI-VALENT DIAMETER INCHES	MINIMUM CORNER RADIUS INCHES	MINIMUM COVER INCHES	2⅔" x ½" CORRUGATIONS					3" x 1" CORRUGATIONS					5" x 1" CORRUGATIONS				
				METAL THICKNESS (INCH/GAGE)														
				0.064/16	0.079/14	0.109/12	0.138/10	0.168/8	0.079/14	0.109/12	0.138/10	0.168/8		0.079/14	0.109/12	0.138/10	0.168/8	
				MAXIMUM FILL HEIGHT ABOVE TOP OF PIPE (FEET)														
17 x 13	15	3	12	13														
21 x 15	18	3	12	12														
24 x 18	21	3	12	13														
28 x 20	24	3	12	13														
35 x 24	30	3	12	12														
42 x 29	36	3.5	12	12														
49 x 33	42	4	12		12													
57 x 38	48	5	12			12												
60 x 46	54	8	15							21					21			
64 x 43	54	6	12			12												
66 x 51	60	9	15							21					21			
71 x 47	60	7	12				12											
73 x 55	66	12	18							20					20			
77 x 52	66	8	12					12										
81 x 59	72	14	18						17					17				
83 x 57	72	9	12					12										
87 x 63	78	14	18						17					17				
95 x 67	84	16	18						17					17				
103 x 71	90	16	18							17				17				
112 x 75	96	18	21							16					16			
117 x 79	102	18	21							16					16			
128 x 83	108	18	24								16					16		
137 x 87	114	18	24								16					16		
142 x 91	120	18	24									16					16	

ALUMINUM

PIPE ARCH SIZE SPAN x RISE INCHES	EQUI-VALENT DIAMETER INCHES	MINIMUM CORNER RADIUS INCHES	MINIMUM COVER INCHES	2⅔" x ½" CORRUGATIONS				3" x 1" CORRUGATIONS			
				METAL THICKNESS (INCH/GAGE)							
				0.060/16	0.075/14	0.105/12	0.135/10	0.060/16	0.075/14	0.105/12	0.135/10
				MAXIMUM FILL HEIGHT ABOVE TOP OF PIPE (FEET)							
17 x 13	15	3	12	13							
21 x 15	18	3	12	12							
24 x 18	21	3	12	13							
28 x 20	24	3	12		13						
35 x 24	30	3	12		12						
42 x 29	36	3.5	15			12					
49 x 33	42	4	15			12					
57 x 38	48	5	15				12				
60 x 46	54	8	15					21			
64 x 43	54	6	18				12				
66 x 51	60	9	18					21			
73 x 55	66	12	18						20		
81 x 59	72	14	21							17	
87 x 63	78	14	21							17	
95 x 67	84	16	24							17	
103 x 71	90	16	24								17

REGISTERED PROFESSIONAL ENGINEER
CODY A. BURCH
PE-42098
DATE 08/27/2026
NORTH DAKOTA

NO SCALE

U.S. DEPARTMENT OF TRANSPORTATION, FHWA OFFICE OF FEDERAL LANDS HIGHWAY	FLH STANDARD 602-1
METAL PIPE CULVERT	SPECIFICATION FP-24, FP-14
	APPROVED FOR USE 8/2024

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29 July 2026 10:50 AM

PROJECT	SHEET NUMBER
ND OMAD 2025(1) MINUTEMAN MISSILE ACCESS ROADS	T2

COUPLING BANDS FOR METAL PIPE CULVERT [1]

CORRUGATION SIZE [2] INCHES	ROUND PIPE DIAMETER INCHES	PIPE ARCH SPAN × RISE INCHES	MINIMUM BAND WIDTH (INCHES)		
			ANNULAR CORRUGATED BANDS [3]	HELICALLY CORRUGATED BANDS [4]	SEMI-CORRUGATED BANDS [5]
1½ × ¼	underline [6]	-	10.5	7	10.5
2⅔ × ½	12 to 36	17 × 13 to 42 × 29	7	12	
	42 to 72	49 × 33 to 83 × 57	10.5	12	
	78 to 84	-	10.5	12	10.5
3 × 1	36 to 72	60 × 46 to 81 × 59	12	14	10.5
	78 to 144	87 × 64 to 142 × 91	12	14	10.5
5 × 1	36 to 72	60 × 46 to 81 × 59	20	22	
	78 to 144	87 × 64 to 142 × 91	20	22	

[1] Fabricate annular, helical and semi-corrugated type coupling bands from the same metal as the connecting pipe. Provide coupling bands not more than 3 nominal sheet thicknesses thinner than the thickness of the pipe to be connected, and no thinner than 0.052 inch for steel or 0.048 inch for aluminum. Fasten coupling bands with the following diameter of bolt:
⅜" for 18" round culvert (21" × 15" pipe arch) or less
½" for 21" round culvert (24" × 18" pipe arch) or more

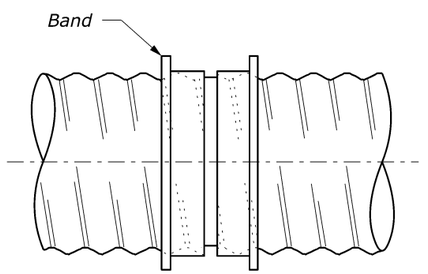
[2] For helically corrugated pipe with rerolled ends, the nominal corrugations size refers to the dimension of the end corrugation in the pipe.

[3] Use annular corrugated bands with pipes having annular corrugations or with helical pipe having rerolled end to form annular corrugations. A 10.5 inch band is acceptable on pipe ends rerolled with 2⅔" × ½" corrugations. A 12 inch band is acceptable on pipe ends rerolled with 3" × 1" pipe corrugations.

[4] Use helical corrugated bands with pipes having helically corrugated ends.

[5] The minimum band widths shown for 3" × 1" and 5" × 1" corrugated sizes apply to 2⅔" × ½" corrugations on rerolled pipe ends.

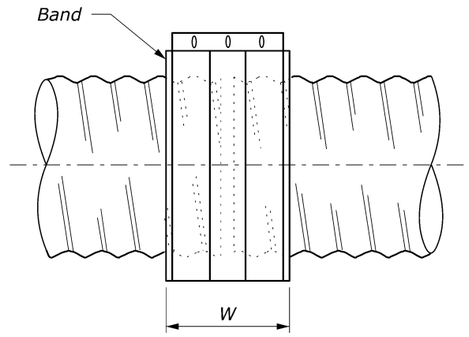
[6] Smooth sleeve-type couplers and flat bands may be used for pipe diameters of 12" or less. Use a matching metal having a nominal thickness of not less than 0.040 inch for steel, or 0.036 inch for aluminum, or a plastic with an equivalent strength to metal.



SLEEVE JOINT

Smoother sleeve with center stop.
Stab type joint

SMOOTH SLEEVE BAND



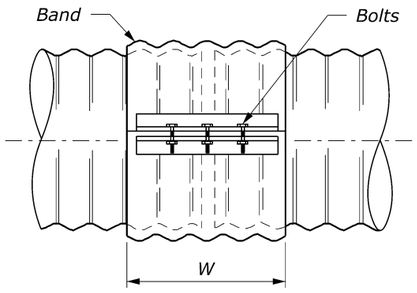
FLAT BAND

NOTE:

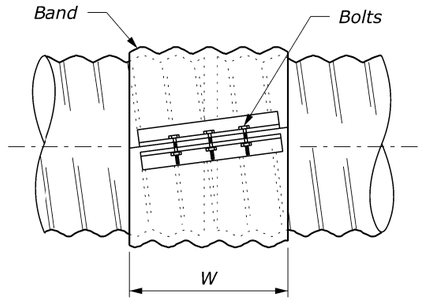
1. Other types of coupling bands or fastening devices that comply with the joint performance criteria of AASHTO Standard specifications for Highway Bridges, Division II Section 26 may be used.

REGISTERED PROFESSIONAL ENGINEER
CODY A. BURCH
PE-42098
DATE 08/27/2026
NORTH DAKOTA

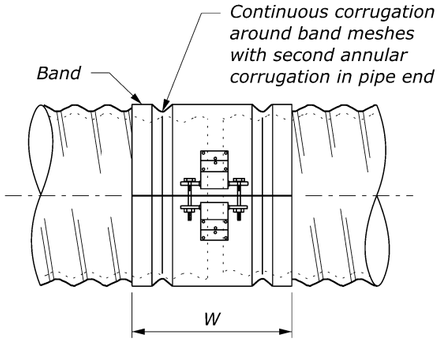
My responsibility with respect to this plan sheet is limited to the selection of the standard plan for this project and a determination that the selection is appropriate for the project



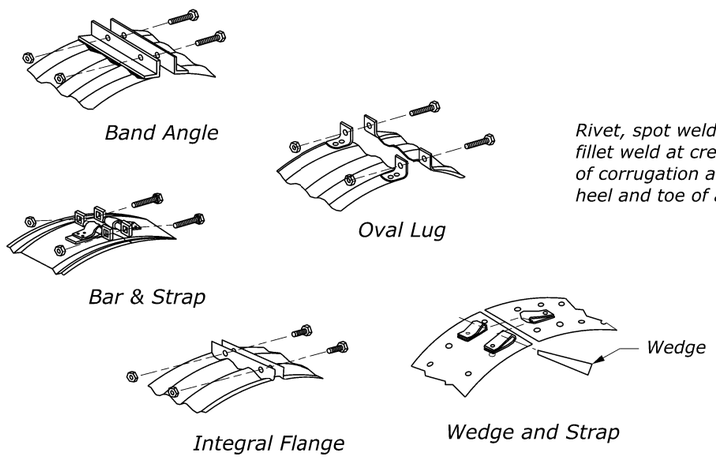
SIDE VIEW



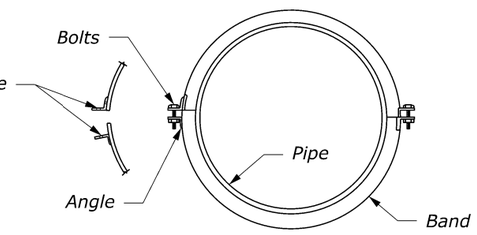
SIDE VIEW



SIDE VIEW



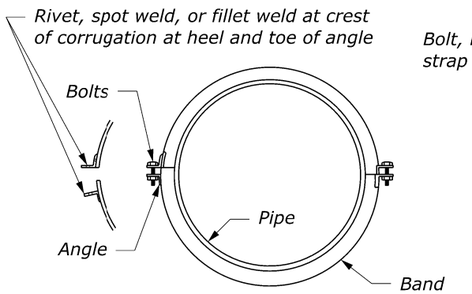
STANDARD BAND CONNECTIONS



END VIEW

Second angle connection optional to 42" diameter, required above 42" diameter

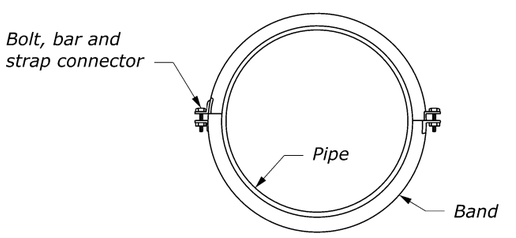
ANNULAR BAND



END VIEW

Second angle connection optional to 42" diameter, required above 42" diameter

HELICAL BAND



END VIEW

SEMI-CORRUGATED BAND

NO SCALE

U.S. DEPARTMENT OF TRANSPORTATION, FHWA OFFICE OF FEDERAL LANDS HIGHWAY	FLH STANDARD 602-2
METAL PIPE CULVERT COUPLING BAND	SPECIFICATION FP-24, FP-14
	APPROVED FOR USE 1/2024

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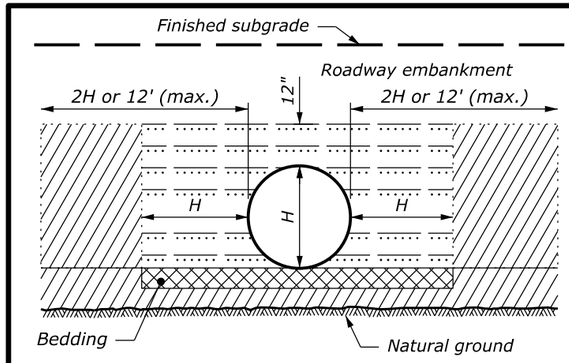
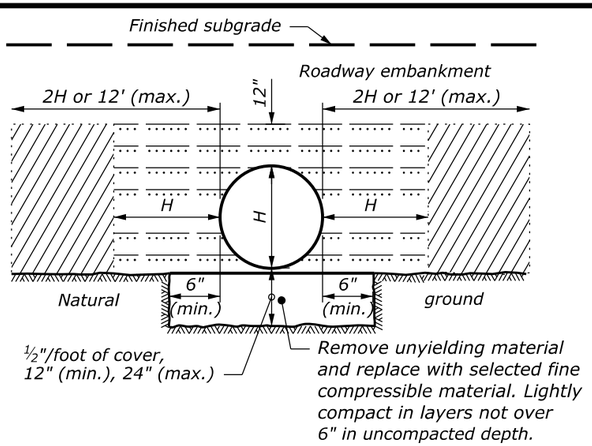
14 August 2026 9:05 AM

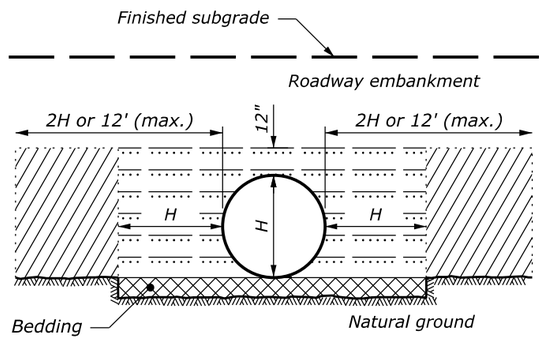
PROJECT	SHEET NUMBER
ND OMAD 2025(1) MINUTEMAN MISSILE ACCESS ROADS	T3



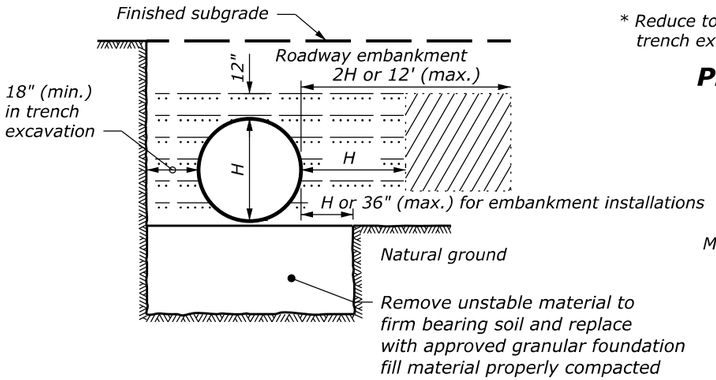
ABOVE NATURAL GROUND



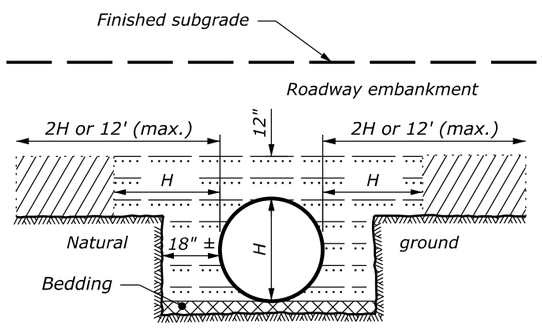
ON UNYIELDING MATERIAL



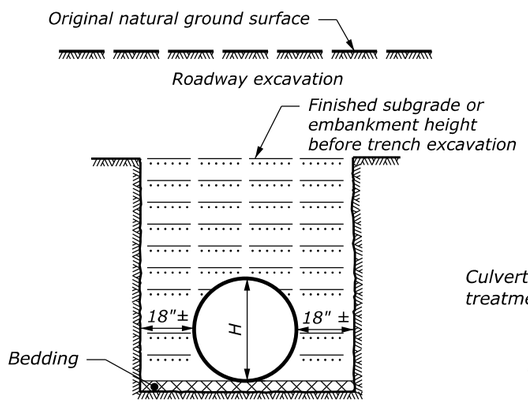
ON NATURAL GROUND



ON UNSTABLE MATERIAL



ABOVE AND BELOW NATURAL GROUND

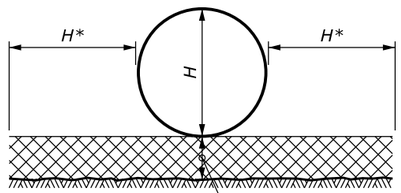


BELOW NATURAL GROUND OR TRENCH EXCAVATION IN EMBANKMENT

LEGEND:

- Bedding material (uncompacted)
- Embankment material placed in layers not exceeding 6" compacted depth.
- Compacted backfill material placed in layers not exceeding 6" compacted depth; or flowable backfill according to Section 614.
- Impermeable backfill material.

BEDDING DEPTH	
PIPE SIZE (H)	DEPTH
12" to 54"	4"
> 54"	6"



PIPE BEDDING

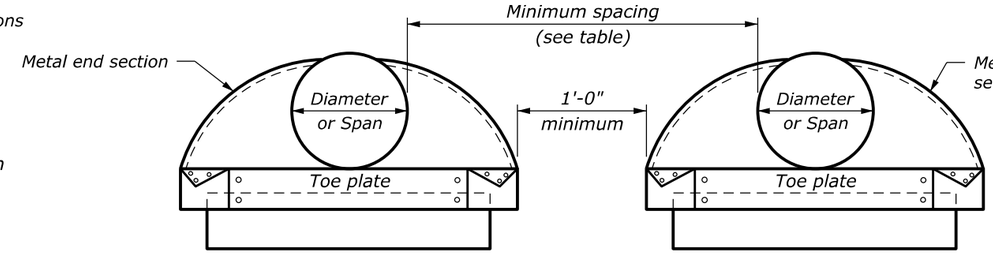
NOTE:

1. When directed, camber pipe culverts upward from a chord through the inlet and outlet inverts an ordinate amount equal to 1% of the pipe length. Develop camber on a parabolic curve. If the midpoint elevation on the parabolic curve as designed exceeds the elevation of the inlet invert, reduce the amount of camber or increase the pipe culvert gradient.
2. H equals the diameter of all round pipe culverts or the rise dimension of all pipe arch culverts.
3. See Section 704 for bedding and backfill requirements.

My responsibility with respect to this plan sheet is limited to the selection of the standard plan for this project and a determination that the selection is appropriate for the project

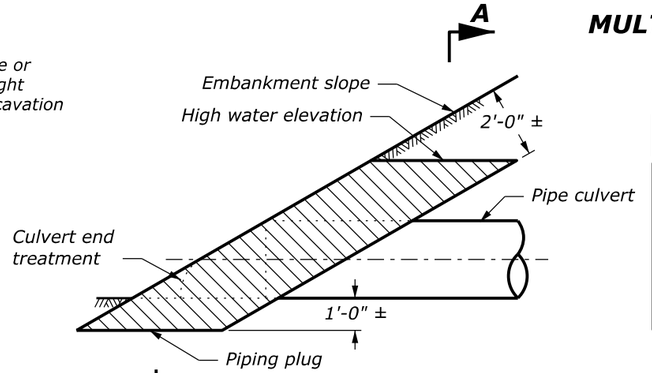
MINIMUM SPACING	
DIAMETER or SPAN	SPACING
UP to 48"	24"
48" and UP	Half diameter or span or 36", whichever is less

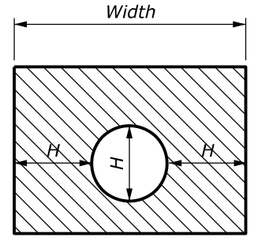
REGISTERED PROFESSIONAL ENGINEER
CODY A. BURCH
PE-42098
DATE 08/27/2026
NORTH DAKOTA



ELEVATION

MULTIPLE PIPE INSTALLATION





SECTION A-A

Construct piping plug of impermeable backfill material at the pipe culvert inlet where granular material is used for backfill. Width may be adjusted to tie into impervious material.

PIPING PLUG

NO SCALE

U.S. DEPARTMENT OF TRANSPORTATION, FHWA
OFFICE OF FEDERAL LANDS HIGHWAY

METAL AND PLASTIC PIPE CULVERT BEDDING

FLH STANDARD	602-3
SPECIFICATION	FP-24
APPROVED FOR USE	1/2024

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14 August 2026 9:40 AM

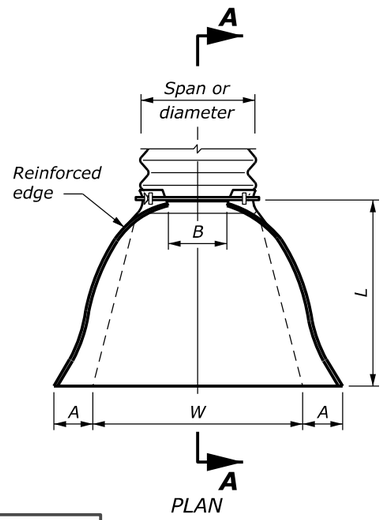
PROJECT	SHEET NUMBER
ND OMAD 2025(1) MINUTEMAN MISSILE ACCESS ROADS	T4

END SECTIONS FOR ROUND PIPE CULVERT

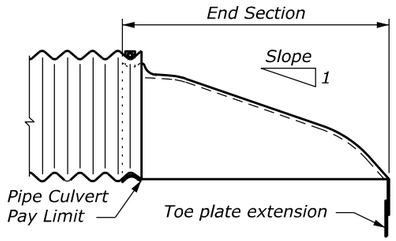
PIPE SIZE DIAMETER INCHES	METAL THICKNESS				DIMENSIONS INCHES					SLOPE Approx.
	STEEL		ALUMINUM							
	INCHES	GAGE	INCHES	GAGE	A (min)	B (max)	H (min)	L (±2")	W (max)	
12	0.064	16	0.060	16	5	7	6	21	44	2¼
15	0.064	16	0.060	16	6	8	6	26	52	2¼
18	0.064	16	0.060	16	7	10	6	31	58	2⅛
21	0.064	16	0.060	16	8	12	6	36	66	2⅛
24	0.064	16	0.060	16	9	13	6	41	72	2⅛
30	0.079	14	0.075	14	11	16	8	51	88	2⅛
36	0.079	14	0.075	14	13	19	9	60	105	2
42	0.109	12	0.105	12	15	25	10	69	122	2⅛
48	0.109	12	0.105	12	17	29	12	78	131	2
54	0.109	12	0.105	12	17	33	12	84	143	2
60	0.109	12	0.105	12	17	36	12	87	157	1⅞
66	0.109	12	0.105	12	17	39	12	87	162	1⅝
72	0.109	12	0.105	12	17	44	12	87	169	1½
78	0.109	12	0.105	12	17	48	12	87	178	1⅜
84	0.109	12	0.105	12	17	52	12	87	184	1⅓
90	0.109	12	0.105	12	17	58	12	87	188	1¼
96	0.109	12	0.105	12	17	58	12	87	197	1⅛

END SECTIONS FOR PIPE ARCH CULVERT

PIPE SIZE SPAN × RISE INCHES	EQUI-VALENT DIAM. (INCHES)	METAL THICKNESS				DIMENSIONS INCHES					SLOPE Approx.
		STEEL		ALUMINUM							
		INCHES	GAGE	INCHES	GAGE	A (min)	B (max)	H (min)	L (±2")	W (max)	
17 × 13	15	0.064	16	0.060	16	7	9	6	19	30	2½
21 × 15	18	0.064	16	0.060	16	7	10	6	23	36	2½
24 × 18	21	0.064	16	0.060	16	8	12	6	28	42	2½
28 × 20	24	0.064	16	0.060	16	9	14	6	32	48	2½
35 × 24	30	0.079	14	0.075	14	10	16	8	39	60	2½
42 × 29	36	0.079	14	0.075	14	12	18	9	46	75	2½
49 × 33	42	0.109	12	0.105	12	13	21	12	53	85	2½
57 × 38	48	0.109	12	0.105	12	18	26	12	63	90	2½
60 × 46	54	0.109	12	0.105	12	18	34	12	70	102	2
64 × 43	54	0.109	12	0.105	12	18	30	12	70	102	2
66 × 51	60	0.109	12	0.105	12	18	33	12	77	116	1½
71 × 47	60	0.109	12	0.105	12	18	33	12	77	114	1½
73 × 55	66	0.109	12	0.105	12	18	36	12	77	126	1½
77 × 52	66	0.109	12	0.105	12	18	36	12	77	126	1½
81 × 59	72	0.109	12	0.105	12	18	39	12	77	138	1½
83 × 57	72	0.109	12	0.105	12	18	39	12	77	138	1½
87 × 63	78	0.109	12	0.105	12	20	38	12	77	148	1½
95 × 67	84	0.109	12	0.105	12	20	34	12	87	162	1½
103 × 71	90	0.109	12	0.105	12	20	38	12	87	174	1½
112 × 75	96	0.109	12	0.105	12	20	40	12	87	174	1½



PLAN



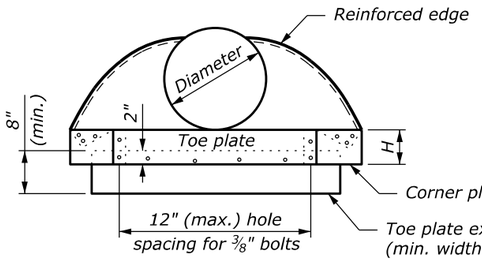
SECTION A-A

ROUND OR PIPE ARCH CULVERT

NOTE:

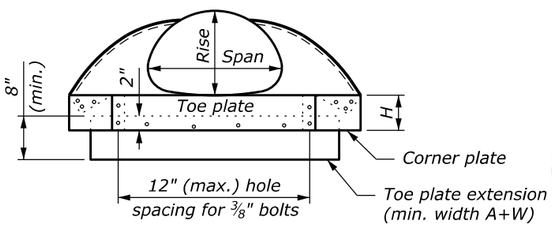
1. Variations in design and dimensions are permitted to allow for manufacturer's standards.
2. Fabricate the diameter of the end section of Design B to match the inside diameter of the concrete pipe culvert.
3. Design C may be used in lieu of design A for all metal pipe culvert sizes. Coupling bands may be any acceptable type for the pipe culvert specified.
4. Fabricate multiple piece bodies with lap seams tightly joined by ⅜" rivets or bolts. Fabricate end section center panels for 60" and larger diameter pipe and equivalent pipe arch from 0.138 inch steel or 0.135 inch aluminum.
5. On end section center panels for 66" and larger equivalent pipe arch provide 2½" × 2½" × ¼" angle reinforcement bolted or riveted under the center panel seam.
6. Supplement the reinforced edges of end sections for 60" and larger diameter pipe and 66" and larger equivalent pipe arch with 2½" × 2½" × ¼" stiffener angles attached with bolts or rivets.
7. Fabricate connector section, corner plate and toe plate extensions from the same metal thickness as the panel body. Use toe plate extension where shown on the plans.
8. Warp embankment slopes to match the slope of the flared end sections.

My responsibility with respect to this plan sheet is limited to the selection of the standard plan for this project and a determination that the selection is appropriate for the project



ELEVATION

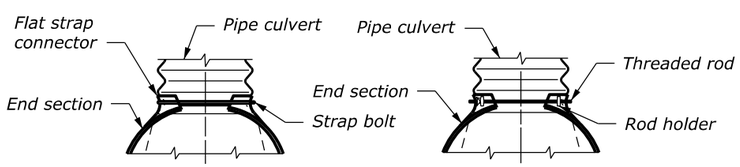
ROUND PIPE CULVERT



ELEVATION

PIPE ARCH CULVERT

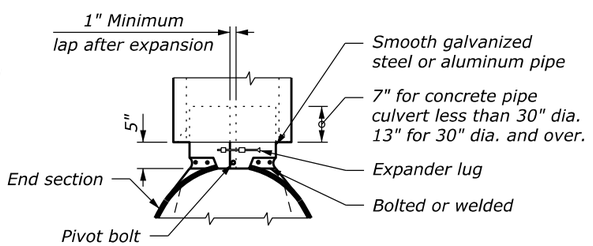
REGISTERED PROFESSIONAL ENGINEER
CODY A. BURCH
PE-42098
DATE 08/27/2026
NORTH DAKOTA



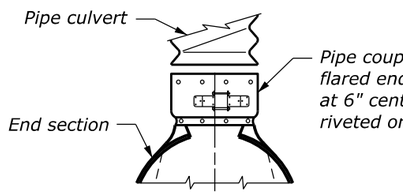
For 12" thru 24" round pipe and 17" × 13" thru 28" × 20" pipe arch

For 30" thru 60" round pipe and 35" × 24" thru 66" × 51" pipe arch

DESIGN A
CONNECTION TO ANNULAR CORRUGATED METAL PIPE



DESIGN B
CONNECTION TO CONCRETE PIPE INLET END



For all sizes of round pipe and pipe arch

DESIGN C
CONNECTION TO METAL PIPE OR OUTLET END OF CONCRETE PIPE

NO SCALE

U.S. DEPARTMENT OF TRANSPORTATION, FHWA OFFICE OF FEDERAL LANDS HIGHWAY	FLH STANDARD 602-4
METAL END SECTIONS	SPECIFICATION FP-24, FP-14
	APPROVED FOR USE 1/2024

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PROJECT	SHEET NUMBER
ND OMAD 2025(1) MINUTEMAN MISSILE ACCESS ROADS	T5

CONCRETE ROUND PIPE CULVERT

FILL HEIGHT AND PIPE CLASS TABLE

PIPE SIZE DIAMETER INCHES	MINIMUM COVER INCHES	EMBANKMENT CLASS II	EMBANKMENT CLASS III	EMBANKMENT CLASS IV	EMBANKMENT CLASS V	TRENCH CLASS II	TRENCH CLASS III	TRENCH CLASS IV	TRENCH CLASS V
		MAXIMUM FILL HEIGHT ABOVE TOP OF PIPE IN FEET							
12	12	10	10	15	23	18	18	26	37
18	12	10	10	25	39	13	13	31	45
24	12	10	10	15	30	15	15	22	40
30	12	9	13	15	35	13	16	20	46
36	12	9	9	20	41	10	13	26	56
48	12	12	13	26	44	15	16	30	49
60	12	15	17	28	44	15	20	32	49
72	12	13	17	30	41	15	20	35	49
84	12	13	19	30		15	23	37	
96	12	13	20			15	24		
108	14	15	20			18	26		

LEGEND:

- Bedding material (uncompacted).
- Embankment material placed in layers not exceeding 6" compacted depth.
- Compacted backfill material placed in layers not exceeding 6" compacted depth, or flowable backfill according to Section 614
- Impermeable backfill material.

REGISTERED PROFESSIONAL ENGINEER
CODY A. BURCH
PE-42098
DATE 08/27/2026
NORTH DAKOTA

NOTE:

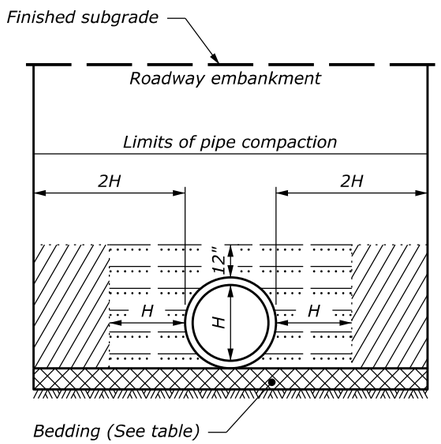
1. When directed, camber pipe culverts upwards from a chord through the inlet and outlet inverts an ordinate amount equal to 1% of the pipe length. Develop camber on a parabolic curve. If the midpoint elevation on the parabolic curve as designed exceeds the elevation of the inlet invert, reduce the amount of camber or increase the pipe culvert gradient.
2. For flexible pavement and aggregate surface roadways, measure minimum cover from the top of the pipe culvert to the bottom of the roadway subgrade. For rigid pavements, measure minimum cover from the top of the pipe culvert to the top of the pavement. For all roadway surface types, measure maximum fill height from the top of the pipe culvert to the top of the pavement.

   Pipe compaction limits shown are for pipe installation in an embankment. For pipe installation in a trench, the compaction limits are the walls of the trench.

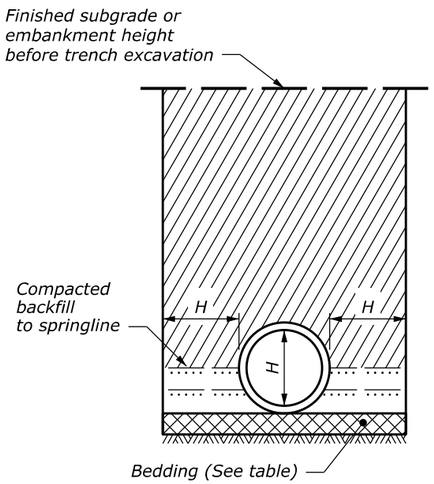
   Where unyielding or unstable material is encountered, install the pipe culvert according to the limits of pipe compaction shown on Standard 602-3.
5. Maximum fill heights for pipe culvert installations may be increased on approval of site-specific structural pipe designs meeting the criteria of AASHTO Standard Specifications for Highway Bridges.
6. Use Supplemental Concrete Pipe Tie when specified in the contract documents.

BEDDING DEPTH	
PIPE SIZE (H)	DEPTH
12" TO 54"	4"
> 54"	6"

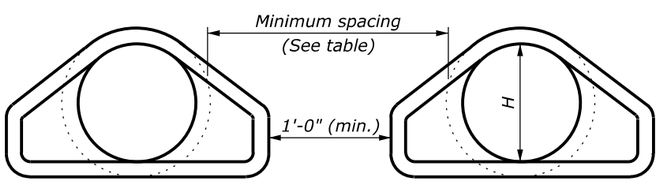
My responsibility with respect to this plan sheet is limited to the selection of the standard plan for this project and a determination that the selection is appropriate for the project



EMBANKMENT INSTALLATION

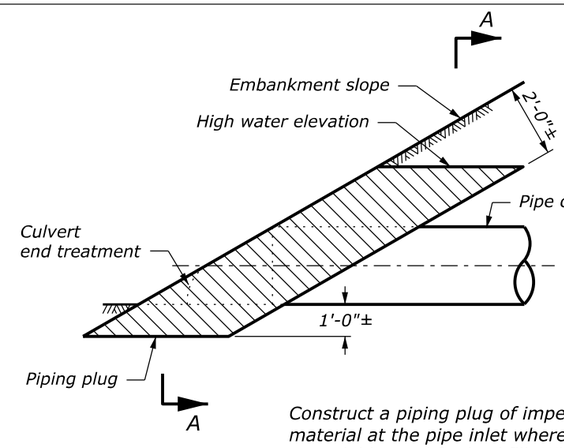


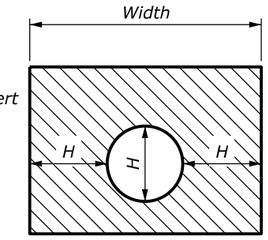
TRENCH INSTALLATION



MINIMUM SPACING		
DIAMETER	EMBANKMENT	TRENCH
12"-36"	15"	2H
36"-96"	0.5H	72"
OVER 96"	48"	72"

MULTIPLE ROUND PIPE INSTALLATION

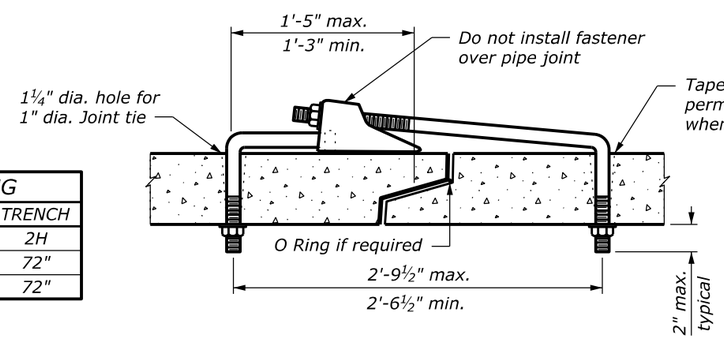




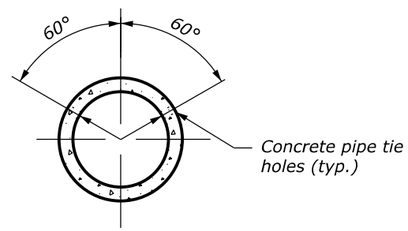
SECTION A-A

Construct a piping plug of impermeable backfill material at the pipe inlet where granular material is used for backfill. Width may be adjusted to tie into impervious material.

PIPING PLUG



SUPPLEMENTAL CONCRETE PIPE TIE



NO SCALE

U.S. DEPARTMENT OF TRANSPORTATION, FHWA
OFFICE OF FEDERAL LANDS HIGHWAY

CONCRETE PIPE CULVERT INSTALLATION

FLH STANDARD	602-7
SPECIFICATION	FP-24
APPROVED FOR USE	8/2024

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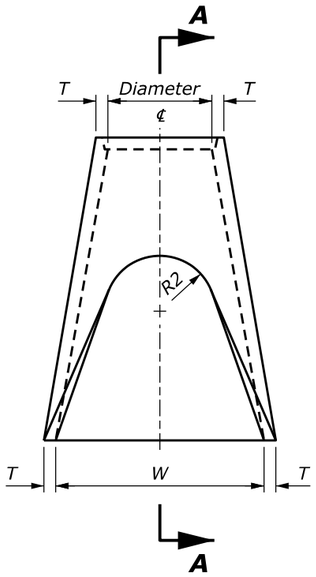
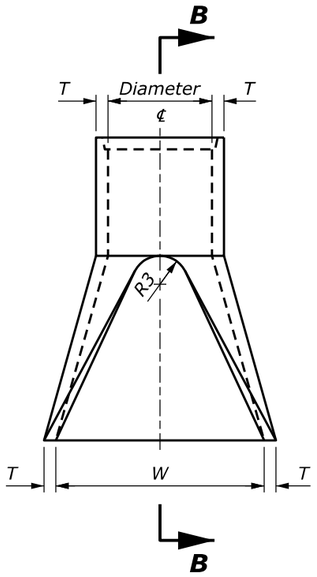
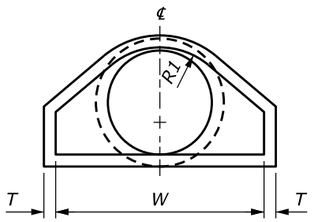
PROJECT	SHEET NUMBER
ND OMAD 2025(1) MINUTEMAN MISSILE ACCESS ROADS	T6



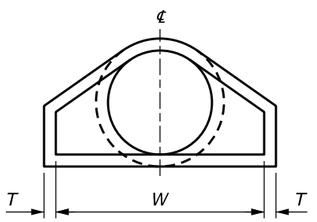
PLAN



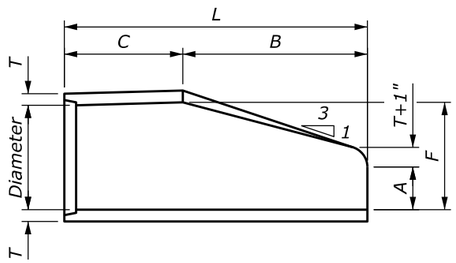
PLAN



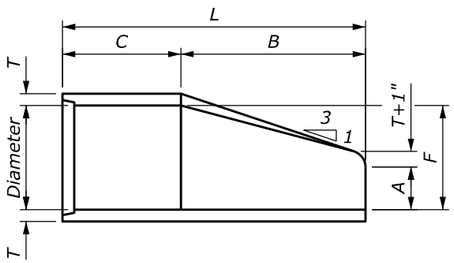
ELEVATION



ELEVATION



SECTION A-A



SECTION A-A

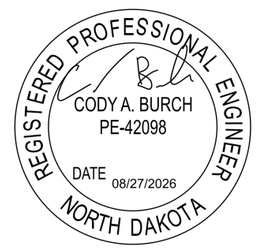
NO SCALE

NOTE:

1. *Variations in design and dimensions are permitted to allow for manufacturer's standards.*
2. *Fabricate the outlet end section with a groove end and the inlet end section with a tongue end.*
3. *Warp embankment slopes to match the slope of the flared end section.*

END SECTIONS FOR ROUND PIPE CULVERT

PIPE SIZE DIAMETER inch	DIMENSIONS inch									
	T	A	B	C	L	W	F	R1	R2	R3
12	2	4	24	48 7/8	72 7/8	24	13	10 1/8	9	4
15	2 1/4	6	27	46	73	30	16	12 1/2	11	6
18	2 1/2	9	27	46	73	36	19	15 1/2	12	7 1/2
21	2 3/4	9	36	37	73	42	22	16 1/2	13	5
24	3	9 1/2	43 1/2	30	73 1/2	48	25	16 3/4	14	8
27	3 1/4	10 1/2	49 1/2	24	73 1/2	54	28	--	14 1/2	9
30	3 1/2	12	54	19 3/4	73 3/4	60	31	18 1/2	15	8
33	3 3/4	13 1/2"	58 1/2	37 1/2	96	66	34	23 3/4	17 1/2	9
36	4	15	63	33	96	72	37	23 1/4	20	11
42	4 1/2	21	63	33	96	78	43	--	22	11
48	5	24	72	24	96	84	49	--	22	12



My responsibility with respect to this plan sheet is limited to the selection of the standard plan for this project and a determination that the selection is appropriate for the project

U.S. DEPARTMENT OF TRANSPORTATION, FHWA
OFFICE OF FEDERAL LANDS HIGHWAY

CONCRETE END SECTION FOR ROUND PIPE

FLH STANDARD	602-8
SPECIFICATION	FP-24, FP-14
APPROVED FOR USE	1/2024

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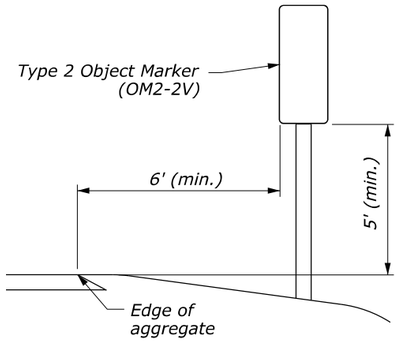
14 August 2026 9:10 AM

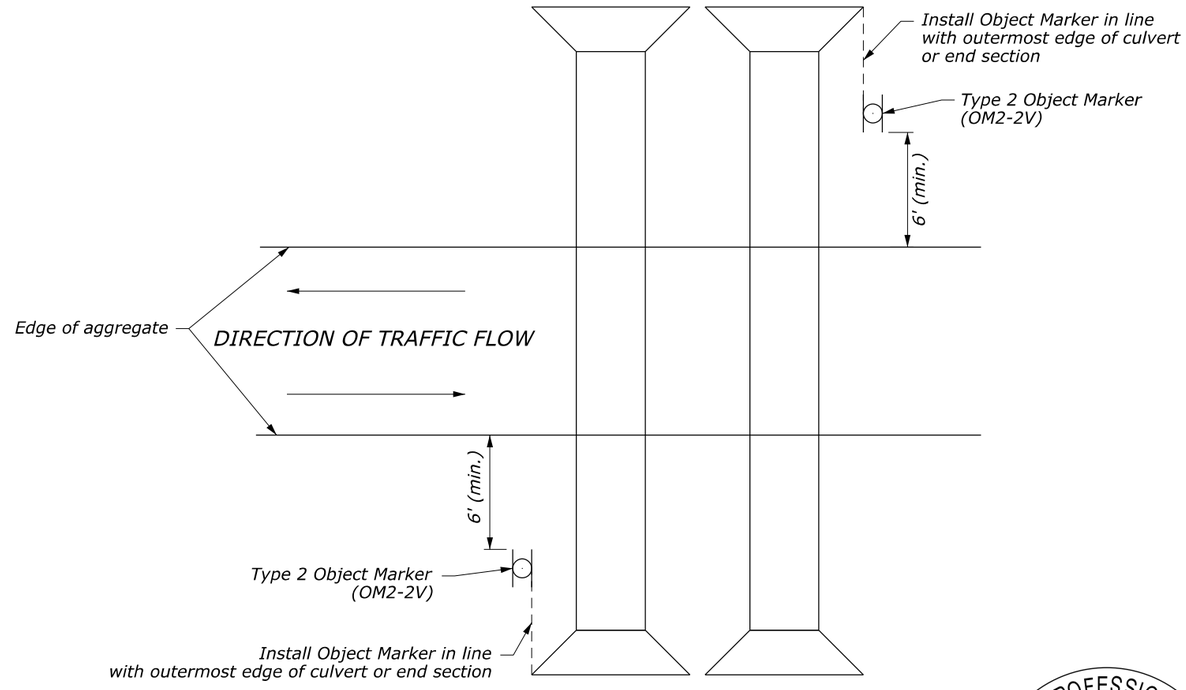
PROJECT	SHEET NUMBER
ND OMAD 2025(1) MINUTEMAN MISSILE ACCESS ROADS	T7



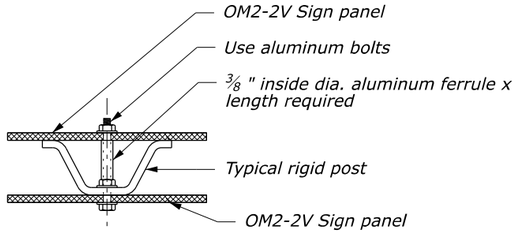
SIGN LOCATION (ELEVATION VIEW)

NOTES:

1. *Locate and set sign height according to the "Manual on Uniform Traffic Control Devices" (MUTCD), latest edition.*
2. *Refer to Section 2A.18 of the MUTCD, latest edition, for additional information.*
3. *Install Type 2 object markers (OM2-2V) using U-channel steel posts according to Section 633. Install two, back to back panels per post.*



SIGN LOCATION (PLAN VIEW)



ATTACHMENT DETAIL

REGISTERED PROFESSIONAL ENGINEER
CODY A. BURCH
PE-42098
DATE 08/27/2026
NORTH DAKOTA

NO SCALE

U.S. DEPARTMENT OF TRANSPORTATION, FHWA OFFICE OF FEDERAL LANDS HIGHWAY	SPECIAL 633-A
OBJECT MARKER DETAIL	

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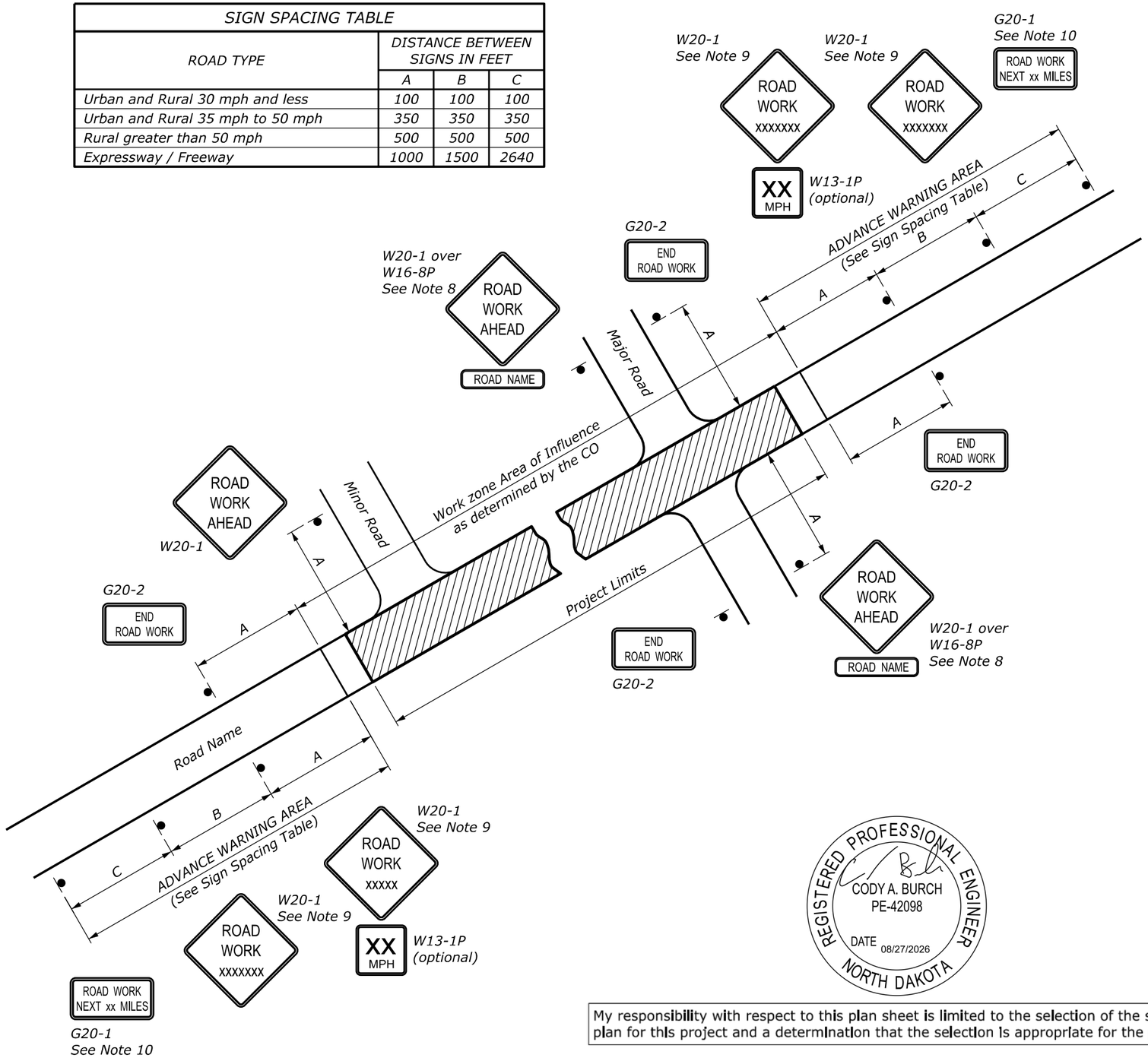
PROJECT	SHEET NUMBER
ND OMAD 2025(1) MINUTEMAN MISSILE ACCESS ROADS	T8

SIGN SPACING TABLE			
ROAD TYPE	DISTANCE BETWEEN SIGNS IN FEET		
	A	B	C
Urban and Rural 30 mph and less	100	100	100
Urban and Rural 35 mph to 50 mph	350	350	350
Rural greater than 50 mph	500	500	500
Expressway / Freeway	1000	1500	2640



NOTE:

1. Erect all project advance warning signs before starting construction work.
2. Not all details shown on the temporary traffic control sheets may be applicable to this project. The Contractor may add or delete information and details in this traffic control plan as necessary to accommodate actual operations.
3. Where advance warning signs, placed as shown, interfere with permanent signs, locate the warning signs to fit field conditions as approved. Vary messages as required.
4. Install advisory speed plaques under the W20 series warning signs as needed to indicate a maximum recommended speed through the construction area.
5. Ensure all sign supports exposed to traffic are crashworthy.
6. Maintain two-way traffic during all non-work hours except as approved.
7. Do not store traffic control devices along the roadway when not in use. Cover post-mounted signs when not applicable.
8. If W20-1 is placed on a roadway other than that on which the actual construction work occurs, include a supplementary plaque indicating the name of the road on which the construction does occur (applies to major roads only).
9. The message on the W20-1 signs may be ROAD WORK AHEAD or may specify the distance to the work area in feet or in miles. Install an additional W20-1 sign when approach speeds exceed 50 mph. When used, place the two W20-1 signs "B" feet apart according to the Sign Spacing Table.
10. For work zones that are 2 miles or more in length, install G20-1 signs at each end of the project. Show the distance on the G20-1 sign to the nearest whole mile.
11. If signing on a roadway under a jurisdiction other than the client agency, verify that an encroachment permit has been obtained.
12. State standards may be used as an alternative if approved.

REGISTERED PROFESSIONAL ENGINEER
CODY A. BURCH
PE-42098
DATE 08/27/2026
NORTH DAKOTA

My responsibility with respect to this plan sheet is limited to the selection of the standard plan for this project and a determination that the selection is appropriate for the project

NO SCALE

U.S. DEPARTMENT OF TRANSPORTATION, FHWA OFFICE OF FEDERAL LANDS HIGHWAY	FLH STANDARD 635-1
TEMPORARY TRAFFIC CONTROL ADVANCE SIGNING	SPECIFICATION FP-24
	APPROVED FOR USE 2/2024

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PROJECT	SHEET NUMBER
ND OMAD 2025(1) MINUTEMAN MISSILE ACCESS ROADS	T9

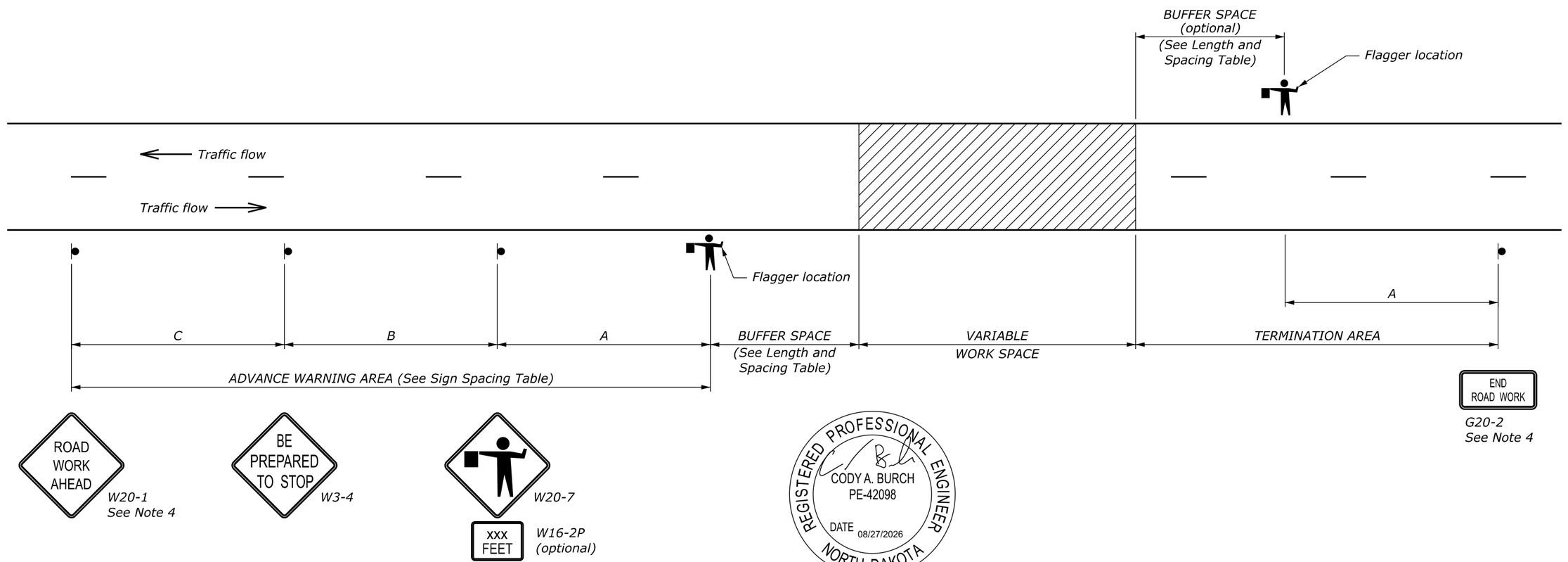
LENGTH AND SPACING TABLE	
APPROACH SPEED* MPH	BUFFER SPACE LENGTH FEET
20	115
25	155
30	200
35	250
40	305
45	360
50	425
55	495
60	570
65	645
70	730

* Approach speed based on the regulatory posted speed, not the advisory speed.

SIGN SPACING TABLE			
ROAD TYPE	DISTANCE BETWEEN SIGNS IN FEET		
	A	B	C
Urban and Rural 30 mph and less	100	100	100
Urban and Rural 35 mph to 50 mph	350	350	350
Rural greater than 50 mph	500	500	500
Expressway / Freeway	1000	1500	2640

NOTE:

1. Signs are shown for one direction of travel only. Place signs similar to those depicted for the opposite direction of travel.
2. Final location and spacing of devices may be changed to fit field conditions as approved.
3. For pilot car operation, mount the PILOT CAR FOLLOW ME (G20-4) sign at a conspicuous location on the rear of vehicle. Prominently display the name of the Contractor on the pilot car.
4. If closure is completely within the project limits, eliminate the ROAD WORK AHEAD (W20-1) and END ROAD WORK (G20-2) signs.
5. Do not allow equipment, materials, or vehicles to be parked or stored in the buffer space.



REGISTERED PROFESSIONAL ENGINEER
CODY A. BURCH
PE-42098
DATE 08/27/2026
NORTH DAKOTA

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NO SCALE

U.S. DEPARTMENT OF TRANSPORTATION, FHWA
OFFICE OF FEDERAL LANDS HIGHWAY

TEMPORARY TRAFFIC CONTROL ROAD CLOSURE LAYOUT (WITH FLAGGERS)

FLH STANDARD	635-5
SPECIFICATION	FP-24, FP-14
APPROVED FOR USE	2/2024

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PROJECT	SHEET NUMBER
ND OMAD 2025(1) MINUTEMAN MISSILE ACCESS ROADS	T10

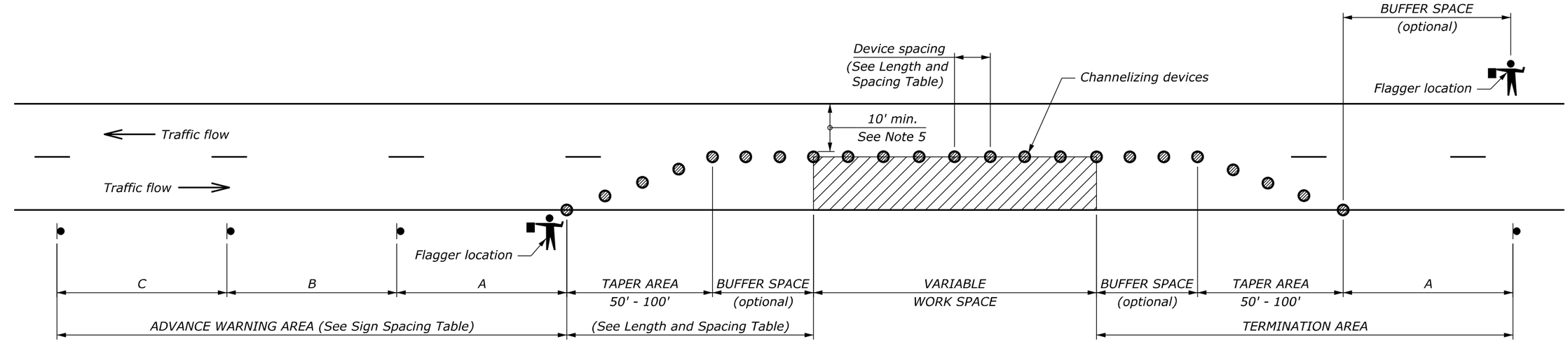
LENGTH AND SPACING TABLE				
APPROACH SPEED* MPH	BUFFER SPACE LENGTH FEET	CHANNELIZING DEVICE SPACING IN FEET		
		TAPER AREA	BUFFER SPACE	WORK SPACE
20	115	20	40	40
25	155	20	50	50
30	200	20	60	60
35	250	20	70	70
40	305	20	80	80
45	360	20	90	90
50	425	20	100	100
55	495	20	110	110
60	570	20	120	120
65	645	20	130	130
70	730	20	140	140

* Approach speed based on the regulatory posted speed, not the advisory speed.

SIGN SPACING TABLE			
ROAD TYPE	DISTANCE BETWEEN SIGNS IN FEET		
	A	B	C
Urban and Rural 30 mph and less	100	100	100
Urban and Rural 35 mph to 50 mph	350	350	350
Rural greater than 50 mph	500	500	500
Expressway / Freeway	1000	1500	2640

NOTE:

1. Signs are shown for one direction of travel only. Place signs similar to those depicted for the opposite direction of travel.
2. Final location and spacing of devices may be changed to fit field conditions as approved.
3. For pilot car operation, mount the PILOT CAR FOLLOW ME (G20-4) sign at a conspicuous location on the rear of vehicle. Prominently display the name of the Contractor on the pilot car.
4. If closure is completely within the project limits, eliminate the ROAD WORK AHEAD (W20-1) and END ROAD WORK (G20-2) signs.
5. For project specific minimum width, refer to the Special Contract Requirements, Section 156.
6. Do not allow equipment, materials, or vehicles to be parked or stored in the buffer space.



W20-1
See Note 4

W20-4

W13-1P
(optional)

W20-7

W16-2P

G20-2
See Note 4

My responsibility with respect to this plan sheet is limited to the selection of the standard plan for this project and a determination that the selection is appropriate for the project

NO SCALE

U.S. DEPARTMENT OF TRANSPORTATION, FHWA
OFFICE OF FEDERAL LANDS HIGHWAY

TEMPORARY TRAFFIC CONTROL SINGLE LANE CLOSURE LAYOUT (WITH FLAGGERS)

FLH STANDARD
635-6
SPECIFICATION
FP-24
APPROVED FOR USE
2/2024

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